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New developments in membranes for bioprocessing – A review

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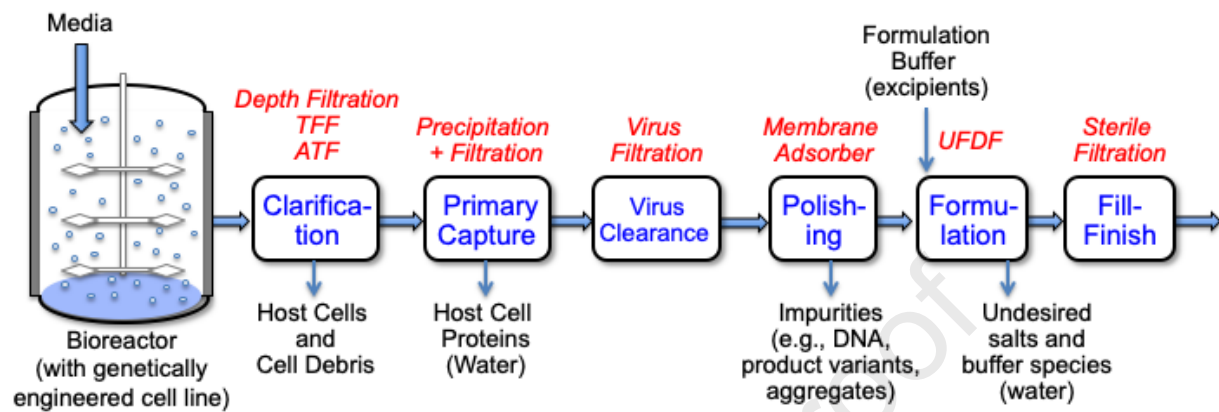
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New Developments in Membranes for Bioprocessing – A Review

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Abstract

Membrane systems play an essential role in the production of all biopharmaceuticals. This includes initial clarification of cell culture fluid by both depth filtration and tangential flow filtration, biomolecule purification by membrane adsorbers, sterilizing grade filtration and bioburden reduction by normal flow filtration, and virus removal by specially designed virus filtration membranes. Recent developments in biomanufacturing are driving new innovations in membrane technology to address the needs associated with continuous bioprocessing, single-use flexible manufacturing, and the production of more complex biotherapeutics for gene and cell-based therapies. This Review examines how the membrane industry has responded to these challenges, with a focus on recent developments in membranes, modules, and processes specifically targeted to biomanufacturing. A brief discussion is also provided of future bioprocessing challenges that will provide unique opportunities for membrane technologies.

Keywords: Bioprocessing, ultrafiltration, microfiltration, antibody, vaccines

Acronyms / Abbreviations

AAV	Adeno-associated virus
ATF	Alternating tangential filtration
BSA	Bovine serum albumin
CAR-T	Chimeric antigen receptor T cell
CCTC	Continuous countercurrent tangential chromatography
CDER	Center for Drug Evaluation and Research
CHO	Chinese hamster ovary
DE	Diatomaceous earth
DF	Diafiltration
DMSO	Dimethyl sulfoxide
Fab	Fragment antigen-binding (for a monoclonal antibody)
Fc	Fragment crystallizable (constant region for a monoclonal antibody)
FDA	Food and Drug Administration
HCP	Host cell proteins
HEK	Human embryonic kidney
hMSC	Human mesenchymal stem cell
HPTFF	High performance tangential flow filtration
ILC	Inline concentrator
ILDF	Inline diafilter
LMH	Unit of L/m ² /h
mAb	Monoclonal antibody
MF	Microfiltration

MLV	Murine leukemia virus
MMV	Mouse minute virus
MWCO	Molecular weight cutoff
PAT	Process analytical technology
PDMS	Polydimethylsiloxane
PES	Polyethersulfone
PVDF	Polyvinylidene fluoride
RO	Reverse osmosis
SPTFF	Single-pass tangential flow filtration
SUS	Single-use systems
TFF	Tangential flow filtration
T-VEC	Talimogene laherparepvec
UF	Ultrafiltration
UFDF	Ultrafiltration / diafiltration
VF	Virus filtration

1. Introduction

The modern biotechnology industry traces its origin to the commercial development of recombinant human insulin [1]. The genes for the A- and B-chains of human insulin were successfully cloned into *E. coli* in 1978 by scientists at Genentech, with the recombinant protein produced commercially by Eli Lilly (licensed from Genentech) beginning in 1982. Published data from 2018 showed that 8 of the top 10 best-selling drugs are biologics produced by recombinant DNA technology, each with world-wide sales of more than \$6.7 billion / yr [2]. Humira, a monoclonal antibody developed by Abbvie for treatment of arthritis and plaque psoriasis, generates nearly \$20 billion in annual revenue by itself [2].

Membrane technology has played an essential role in the development of all of these biopharmaceutical products. A process flow diagram for the early insulin process showed 8 distinct membrane steps, which does not even account for the multiple 0.2 μm pore size membranes used to filter various buffers and process streams to remove particulates and reduce bioburden [3]. The platform process that is currently used for production of nearly all monoclonal antibodies employs at least 4 distinct membrane steps [4]:

1. depth filtration for the initial clarification of the cell culture fluid
2. virus filtration for removal of adventitious and exogenous viruses
3. ultrafiltration / diafiltration for concentration, buffer exchange and formulation
4. sterile filtration for bioburden reduction and final fill-finish operations

In their review of membrane separations in biotechnology, Rathore and Shirke [5] noted that typical processes for the production of biopharmaceuticals have anywhere from 10-20 membrane-based separation steps.

Figure 1 shows a general schematic for the production of extracellular (secreted) bioproducts. Membrane technology is used throughout the process. Although clarification of the cell culture fluid is commonly performed by centrifugation, this is typically augmented by depth filtration and / or normal flow microfiltration to achieve the final clarity needed to minimize fouling in subsequent chromatography steps [6]. In addition, tangential flow filtration (TFF), and its variant known as alternating tangential filtration (ATF), are the dominant clarification technologies for perfusion bioreactors which are designed to provide continuous harvesting of the secreted product with recycle of the genetically engineered cells back to the bioreactor. Primary capture of monoclonal antibody products is typically accomplished using Protein A chromatography, but there is growing interest in the use of precipitation / flocculation with the product then recovered by membrane filtration. Virus filtration is an essential part of the overall strategy for viral clearance, complimenting virus inactivation (e.g., use of low pH hold or various solvents / detergents) and adsorptive removal (e.g., by ion exchange chromatography) to ensure the safety of the biopharmaceutical product, particularly for plasma proteins and mammalian-cell derived products. Polishing operations are commonly performed using column chromatography, but a variety of membrane adsorbers have been developed that can provide enhanced performance due to high convective flow rates. In addition, tangential flow filtration systems have been developed for specific purification steps in the production of glycoconjugate vaccines. Ultrafiltration and diafiltration (UFDF) are used for the final formulation of essentially all biotherapeutics, providing both concentration and buffer exchange / desalting in a single unit operation. Sterilizing grade filtration is an integral part of the fill-finish operations, ensuring that the final product is completely free of any microbial contamination.

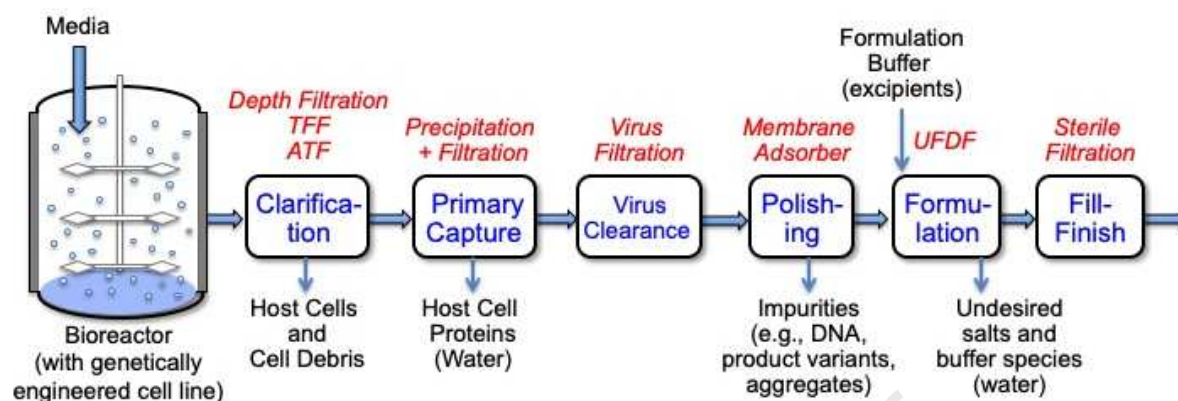


Figure 1 – Schematic of a typical process flow diagram for production of a biopharmaceutical product.

van Reis and Zydney [7] provided a very thorough discussion of the use of membrane processes in the biotechnology industry in their 2007 review paper in the *Journal of Membrane Science*. A number of subsequent reviews have appeared since then, most notably the review by Rathore and Shirke in 2011 [5] and the book *Ultrafiltration for Bioprocessing* by Herb Lutz in 2015 [8]. The bioprocessing industry has continued to evolve in the years since these reviews were published, driven by several distinct trends in biomanufacturing, most particularly:

- application of single-use (disposable) systems that facilitate the development of multi-product manufacturing facilities [9]
- development of continuous / connected processes with enhanced flexibility, reduced cost, and greater product quality [10]
- production of larger and more complex biotherapeutics, including conjugated vaccines, virus-like particles, gene therapy agents, and cell-based therapeutics

The objective of this Review is to highlight how the membrane industry has responded to these bioprocessing trends through the development of new membranes, modules, and processes that are specifically designed to meet the evolving needs of the biotechnology industry.

2. Single use systems

Single-use (or disposable) systems for bioprocessing are typically supplied in ready-to-use format, pre-sterilized using either gamma irradiation or ethylene oxide gas [11]. This eliminates the need for subsequent steam sterilization and cleaning, significantly reducing energy costs, water usage, and the disposal of environmentally challenging cleaning chemicals. Capital costs for single-use systems are significantly lower than those for stainless steel installations, and these facilities can also be brought on-line more rapidly to reduce manufacturing timelines. In addition, the adoption of single-use technology greatly facilitates change-over in multiproduct facilities, minimizing turn-around time while significantly reducing the risk of cross-contamination. This greatly enhances manufacturing flexibility, improving the industry's ability to respond to fluctuations in demand [12].

Single-use systems (SUS) currently make up about 85% of the equipment employed in the production of pre-clinical materials, and these systems are increasingly being used to support clinical trials and for larger-scale commercial biomanufacturing [12]. The growth in the market for single-use systems in bioprocessing is expected to reach \$5.44 billion by 2021 with a projected annualized growth rate of nearly 20% [13].

Almost all normal flow filters were originally designed as single use systems, including both sterilizing-grade and virus removal filters. The principles underlying the design of these normal flow filtration processes are described in van Reis and Zydney [7]. The current

discussion is focused on the development of depth filters for clarification of cell cultures, single use systems for tangential flow ultrafiltration, and the application of membrane adsorbers / membrane chromatography as a replacement for large stainless steel chromatography columns.

2.1 Depth Filters

In contrast to membranes, which typically retain species by a surface sieving mechanism, depth filters have more open structures in which retention occurs throughout the entire depth of the pore volume by a combination of size and surface interactions. Depth filtration was originally used in bioprocessing to provide an additional level of clarification after centrifugation. However, depth filtration has now been developed as a stand-alone clarification step, sometimes used in combination with flocculation / precipitation, to remove both cells and cell debris in single-use processes [14]. This typically requires the use of depth filters with graded pore structures or the use of multiple depth filters in series, each with different pore size, to effectively remove the wide size range of the cells and debris present in most mammalian cell culture streams.

Most commercial depth filters are heterogeneous structures consisting of a fiber matrix, a filter aid, and binder (Figure 2). The matrix is typically a non-woven array of cellulose or polypropylene fibers, with the diameter and packing density of these fibers determining the effective pore size of the filter. Smaller debris, proteins, and DNA are primarily removed by the high surface area filter aid contained within the matrix, e.g., diatomaceous earth (DE), activated carbon, or perlite. The binder is used to maintain the physical stability of the overall filter structure, although there is also evidence that the binder can provide significant adsorptive removal of proteins and DNA in some filters.

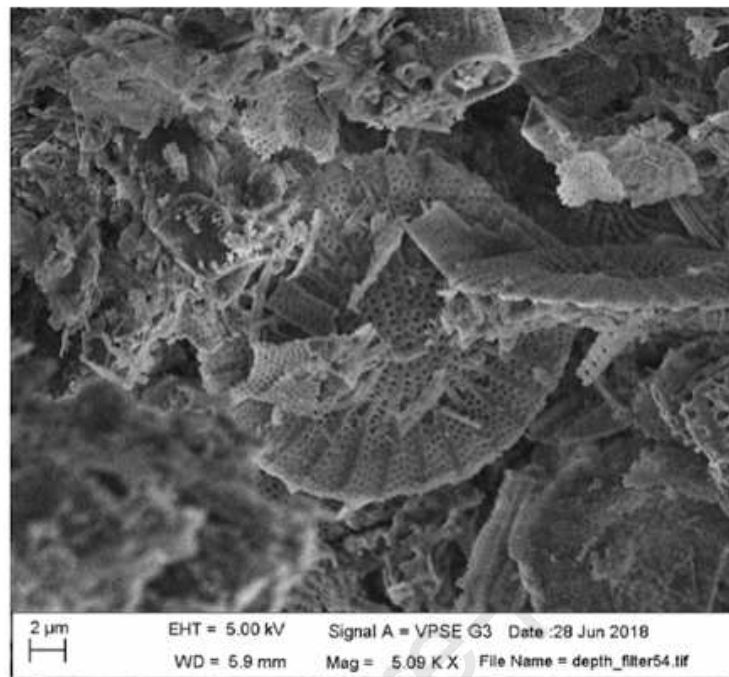


Figure 2 – Scanning electron micrograph of a cross-section through the Pall PDH4 depth filtration media containing diatomaceous earth. From [15].

Small scale depth filters are housed as flat sheets in small sealed capsules (left panel of Figure 3). It is very difficult to effectively pleat most depth filter materials; thus, pilot and commercial scale depth filters typically use a parallel array of these sheets, with the flow distributed among the layers or pads (right panel). The detailed flow distribution in these modules can be quite complex [16], particularly given the very high permeability of the depth filter media. This can create challenges in the scale-up / scale-down of depth filtration processes [17].

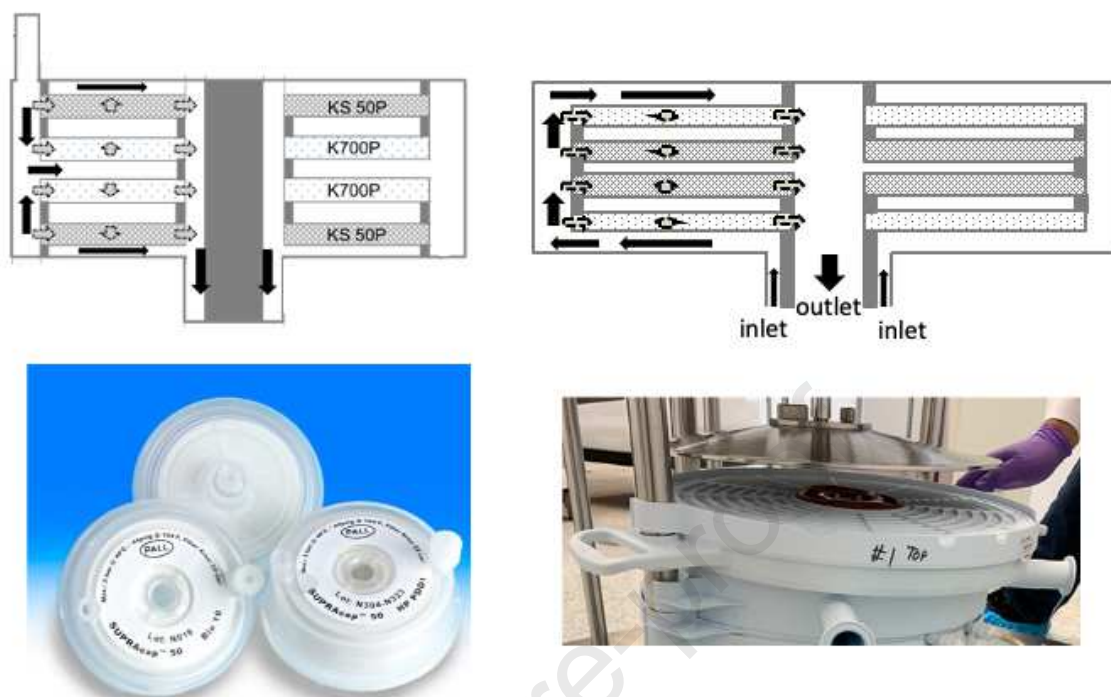


Figure 3 – Geometry and flow distribution in small scale (left panels) and pilot scale (right panels) depth filters from Pall Corp. The K700P media has a 6-15 μm pore size while the KS 50P media has a 0.4 – 0.8 μm pore size. From [16].

Depth filters remove cells and cell debris on both the external surface and within the depth of the filter due to primarily to size-based retention by the porous matrix. For example, Singh et al. [14] showed that steeply graded depth filters, in which the effective pore size decreases by as much as a factor of 10 as one moves from the filter inlet to outlet, can provide very effective removal of turbidity even from mammalian cell culture streams with very high cell density. For example, the ClarisolveTM depth filter from Millipore provided capacities of more than 100 L/m² for an untreated CHO cell culture harvested at 80-90% viability with a cell density of 15-20 x 10⁶ cells / mL [14]. The capacity could be increased by more than 3-fold by treatment with acid precipitation or by flocculation with chitosan or a polymeric flocculant, both of which

reduced the amount of smaller debris. The clarified cell culture fluid could then be processed through a 0.2 μm pore size (sterilizing grade) filter with capacities of $>1000 \text{ L/m}^2$ [14].

Although the primary purpose of the depth filtration step is clarification, several studies have demonstrated that these filters can also provide significant adsorptive removal of host cell proteins, DNA, endotoxins, and viruses, effectively reducing the process demands for the subsequent purification steps. Venkiteshwaran et al. [17] showed that high ionic capacity depth filters could provide >4.8 log-removal of porcine parvovirus when operated at low conductivity. Khanal et al. [19] showed that protein removal by depth filters containing diatomaceous earth or perlite was due primarily to electrostatic interactions with the positively-charged polymeric binder. In contrast, Nejatishheiden et al. [15] found that the negatively-charged diatomaceous earth provided greater removal of positively-charged proteins. A number of studies have shown that hydrophobic interactions can provide a significant contribution to the binding interactions for some impurities [20]. These results clearly demonstrate that the detailed physical chemistry of the depth filter plays a critical role in determining the removal rates for small impurities.

Depth filters can also be used for impurity removal in the production of biotherapeutics produced in bacterial cells. In this case, the initial cell recovery is typically performed by centrifugation or tangential flow filtration, with the washed cells then lysed to release the intracellular product. Metzger et al. [21] demonstrated that the EmphazeTM anion exchange depth filter from 3M could provide 3-log removal of DNA and 7-log removal of endotoxin, along with a small reduction in HCP, significantly reducing the rate of fouling in a subsequent cation exchange chromatography column. Scheffler et al. [22] demonstrated the effectiveness of DE containing depth filters at removing precipitated mAb after a low pH-hold viral inactivation step.

Although depth filtration is an inherently batch operation, with the filter discarded after the maximum capacity is reached, Thakur et al. [23] designed a dead-end filtration skid for continuous bioprocessing in which the feed is switched between parallel depth filters at previously set endpoints for the maximum turbidity or pressure. The effectiveness of this system was demonstrated for the initial clarification of recombinant proteins produced by *Escherichia coli*, *Pichia pastoris*, and Chinese Hamster Ovary (CHO) cell cultures. This general approach could also be used to adapt other normal flow filtration systems to continuous bioprocessing.

2.2 Membrane Adsorbers

Single-use (disposable) chromatography systems are particularly attractive for small scale bioprocessing, both in the production of material for early clinical studies and for manufacturing of products with relatively small annual demand [24]. These systems eliminate the need for column packing operations, reducing labor costs and potentially providing more uniform flow properties than can be obtained even with automated packing systems. This also eliminates the need for column stripping, clean-in-place, and regeneration steps. Not only does this reduce costs, it also minimizes the introduction of bioburden, which can be a particular issue during column packing. SUS chromatography modules can be made using conventional resins, from porous monoliths (typically cylindrical rods or disks of porous material with highly interconnected pore structures), or from stacks of porous membranes.

Membrane chromatography modules, often referred to as membrane adsorbers, are ideal for single-use applications [25]. These systems employ large pore size membranes that contain functionalized polymers that provide purification based on affinity, ion exchange, and / or hydrophobic interactions. For example, the NatriFlo® membrane chromatography technology

(originally developed by Natrix Separations and now sold by MilliporeSigma) uses a porous polyacrylamide hydrogel with a high density of binding ligands reinforced by an inert fiber matrix (Figure 4). Effective pore size is approximately 0.4 μm , providing high surface area for adsorption. Transport through the porous membrane is controlled by convection, with much lower mass transfer resistance than packed columns. This enables the use of higher flow rates at lower pressure drops, which is particularly attractive for disposable (plastic) systems. The required bed volume is typically achieved by using multiple membranes directly on top of each other (in a “stack”) or as part of a multi-layer pleated cartridge.

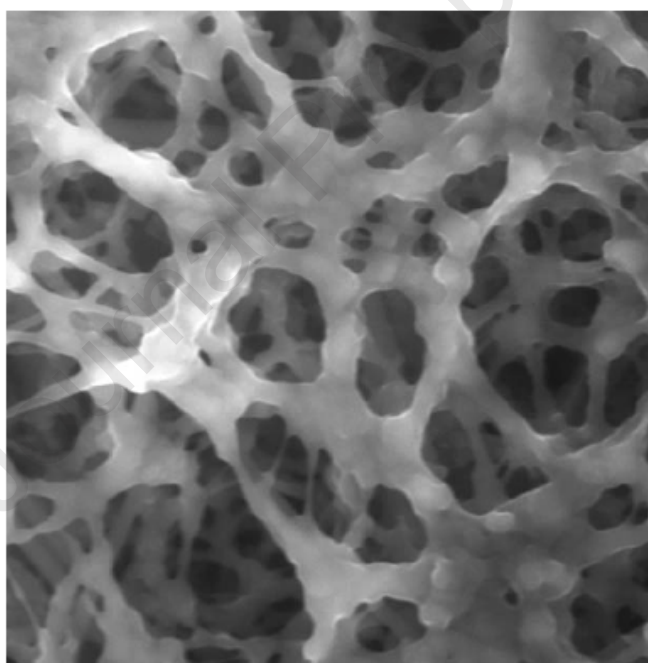


Figure 4: Scanning electron micrograph of the NatriFlo® membrane chromatography media consisting of a functionalized polyacrylamide hydrogel supported by an inert matrix. From [25].

One approach that has been used to enhance the performance of membrane adsorber technology is the laterally-fed membrane chromatography device (Figure 5) developed by Raja

Ghosh and co-workers [26, 27]. This device uses a lateral flow channel to more evenly distribute the flow over the feed-side of a stack of flat sheet membranes, with a channel on the permeate side providing a similar pressure drop to maintain a uniform transmembrane pressure across the adsorber. Data obtained with mixtures of model protein using Sartobind S cation exchange membranes showed better resolution than that for a packed column containing Sepharose SP resin [27].

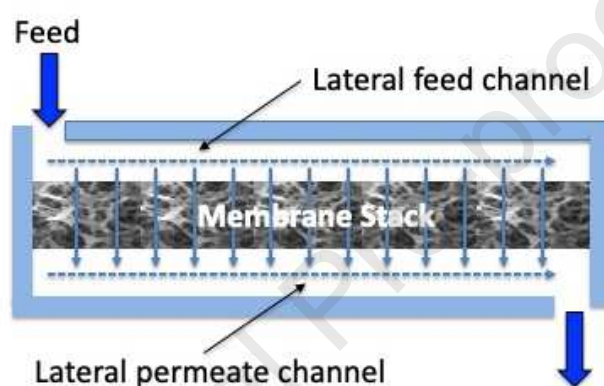


Figure 5: Laterally-fed membrane chromatography device with uniform flow distribution across membrane stack.

The most common use of membrane adsorbers in bioprocessing is for flow through chromatography in which the impurities are bound to the membrane while the product passes freely through the porous structure. Commercial anion exchange membranes with quaternary amine functionality are available from Sartorius (Sartobind), Pall (Mustang), MilliporeSigma (NatriFlo), and Cytiva (FibroSelect), among others. These membranes typically have DNA binding capacities greater than 20 mg/mL and can also provide significant endotoxin removal [28]. Masuda et al. [29] showed that the NatriFlo® HD-Q provided approximately 5-log removal of both mouse minute virus (MMV) and murine leukemia virus (MLV) from a monoclonal

antibody product, making these membrane adsorbers very attractive as part of the overall viral clearance strategy in the purification of mammalian cell-derived products.

More recently, a number of companies have developed synthetic “hybrid” filters that combine many of the features of depth filters and membrane adsorbers. This includes the Emphaze™ AEX and ST-AEX (anion exchange) materials developed by 3M as well specially designed synthetic prefilters for use in combination with virus filtration, e.g., the Viresolve® Pro Shield (cation exchange chemistry) and Shield-H (mixed mode chemistry) from MilliporeSigma. Singh et al. [30] successfully demonstrated the effectiveness of the Emphaze™ ST-AEX, a multilayer filter containing a non-woven polypropylene with a quarternary amine functionality followed by a guanidinium-functionalized polyamide membrane, to replace both a traditional depth filter and a flow through anion exchange step in the post-Protein A purification of several mAbs and a fusion protein. Tang et al. [31] showed that the Viresolve® Pro Shield provided more than a 20-fold increase in capacity during filtration of heavily-fouling mAbs through the Viresolve® Pro virus filtration membrane.

There have also been major efforts to increase the capacity of membrane adsorbers so that they can be effectively used in bind-and-elute mode, e.g., for initial capture of monoclonal antibody products. This includes the use of immobilized hydrogels supported by an inert framework [25], grafting of polymer brushes onto polypropylene membranes [32], and use of surface modified electrospun nanofibers [33].

Protein A affinity membranes are now available with similar capacities as Protein A resins, but these are typically too expensive for single use applications. Bramer et al. [34] examined the use of the Sartobind® Protein A membrane adsorber for mAb purification, including the development of a “multi-adsorber” process that could be employed for continuous

chromatographic separations. Multimodal membranes are more attractive for single use capture. For example, the NatriFlo® HD multimodal membrane with sulfonic acid and t-butyl (hydrophobic) ligands has greater than 90 g/L dynamic binding capacity (evaluated at 10% breakthrough) for positively-charged monoclonal antibodies [28]. Wang et al. [35] developed a multi-modal membrane using atom transfer radical polymerization (ATRP) that has an equilibrium (static) binding capacity for human serum IgG of more than 100 g/L with similar levels of host cell protein removal as provided by a Protein A resin during the initial capture of a CHO-derived monoclonal antibody.

2.3 Single-use TFF

Ultrafiltration / diafiltration (UFDF) is used for final concentration and buffer exchange in the formulation of nearly all biopharmaceutical products [7]. These units employ membranes with pore size between about 1 and 5 nm, providing high retention of the biopharmaceutical and relatively free passage of lower molecular weight impurities. Membranes are typically made from polysulfone, polyethersulfone, or cellulose. The cellulose membranes are more hydrophilic with lower levels of protein adsorption and fouling, although cellulose is less stable to harsh cleaning chemicals [7]. The latter is of no concern with single-use systems.

Ultrafiltration modules are operated as tangential flow filtration (TFF), sometimes referred to as crossflow filtration, with the feed flow parallel to the membrane surface and perpendicular to the direction of filtration [7]. The tangential flow sweeps the membrane surface, enhancing back mass transport into the bulk, reducing fouling and providing much higher filtrate flux than can be achieved with normal flow filtration given the high concentration of retained product. Traditional ultrafiltration systems consist of large stainless steel skids, with the

membrane modules fed from large stainless steel tanks. van Reis and Zydney [7] provide a photograph of a typical system with 250 m² membrane area at Genentech. In contrast, single use systems are typically fed from disposable plastic bags, with volumes up to 2000 L, and have fully disposable flow paths made from appropriate plastic tubing and sterile connectors. These systems typically incorporate single use sensors for pressure, flow, and temperature, and possibly conductivity and UV absorbance.

Magarian et al. [36] demonstrated that UFDF can also be used to remove a wide range of extractables / leachables that might be generated from single-use bags and other systems used throughout the biomanufacturing process. Data obtained with acetonitrile, benzyl alcohol, t-butyl alcohol, butyrolactone, caprolactam, hexanoic acid, and trimethylsilanol showed 1000-fold reduction in concentration of these model extractables / leachables within 9 diavolumes (N_D), consistent with standard models for solute clearance during diafiltration [7]:

$$\frac{c}{c_o} = \exp(-S_o N_D) \quad (1)$$

where N_D is the ratio of the cumulative filtrate volume to the feed (or retentate) volume during the constant volume diafiltration process and S_o is the observed sieving coefficient, equal to the ratio of the solute concentration in the permeate to that in the retentate solution; S_o will be nearly equal to one for small extractables and leachables. Magarian et al. [36] did observe a rebound phenomenon, in which the solute concentration increased during the protein recovery step used at the end of the UFDF process. This recovery step was conducted with reverse flow (from retentate outlet to feed inlet) to re-suspend any product that might be adhering to the membrane and / or within the concentrated boundary layer adjacent to the membrane surface.

Figure 6 shows a comparison of the processing times for a typical single use and re-usable system for final formulation of a recombinant protein product by UFDF. The single-use

systems are pre-sanitized using gamma irradiation, allowing them to be used without an initial sanitization step (including the subsequent flushing to remove chemicals used for sanitization). The single use systems do not require an extensive post-use cleaning cycle to regenerate the membrane; residual product can simply be recovered by a buffer flush. This also means that there is no need to evaluate the normalized water permeability after the UFDF process, which is typically done as part of the cleaning cycle to validate that the membrane was successfully regenerated [37]. The net result is more than a 30% reduction in total process time.

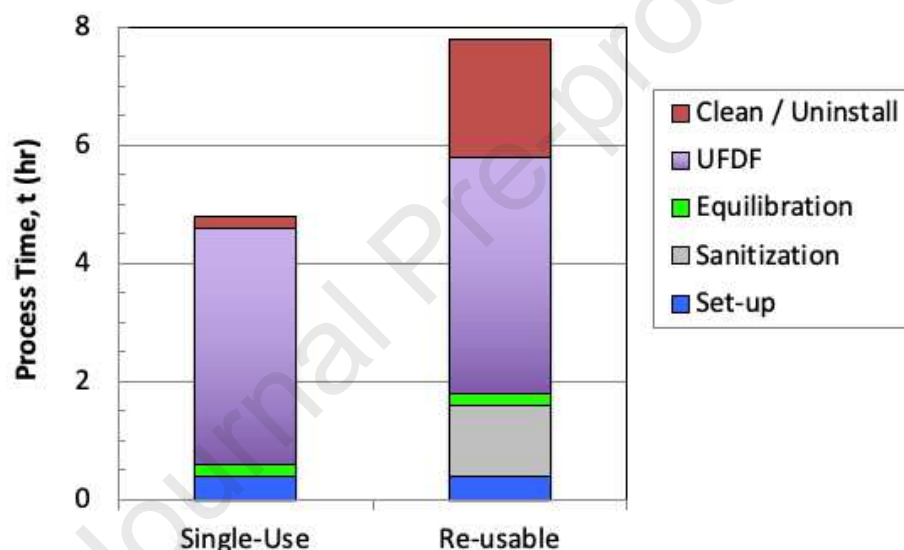


Figure 5 – Comparison of typical process times for single-use and re-usable UFDF process.

A number of pre-sterilized (typically gamma-irradiated) single-use TFF modules have been developed specifically for bioprocessing applications, including the AllegroTM single-use system (originally from Pall, now sold by Sartorius), the Mobius® FlexReady Solution for tangential flow filtration (from MilliporeSigma), and the TangenXTM SIUSTM single use cassette (originally from Novasep, now sold by Repligen). More recently, MilliporeSigma has developed

the Pellicon® Capsule, a holder-less single use TFF device that is gamma sterilized and provided in preservative-free RO water with aseptic connectors, greatly facilitating their use as a “plug-and-play” technology in bioprocessing applications. Single use hollow fiber modules are available from Cytiva (formerly GE Healthcare). These modules are designed to mimic the corresponding re-usable systems, with comparable feed-side pressure drops (ΔP) and mass transfer coefficients (k_m), but with aseptic connectors to facilitate their application in single-use systems. This is accomplished by using equivalent flow geometries (e.g., length and channel height or fiber radius), including similar spacer elements (screens) within the cassettes (Figure 7). This ensures that the modules have essentially the same filtrate flux (J_v) for a given set of feed conditions:

$$J_v = k_m \ln \left(\frac{C_w}{C_b} \right) \quad (2)$$

where C_w and C_b are the protein concentrations immediately upstream of the membrane surface (the “wall”) and in the bulk solution.

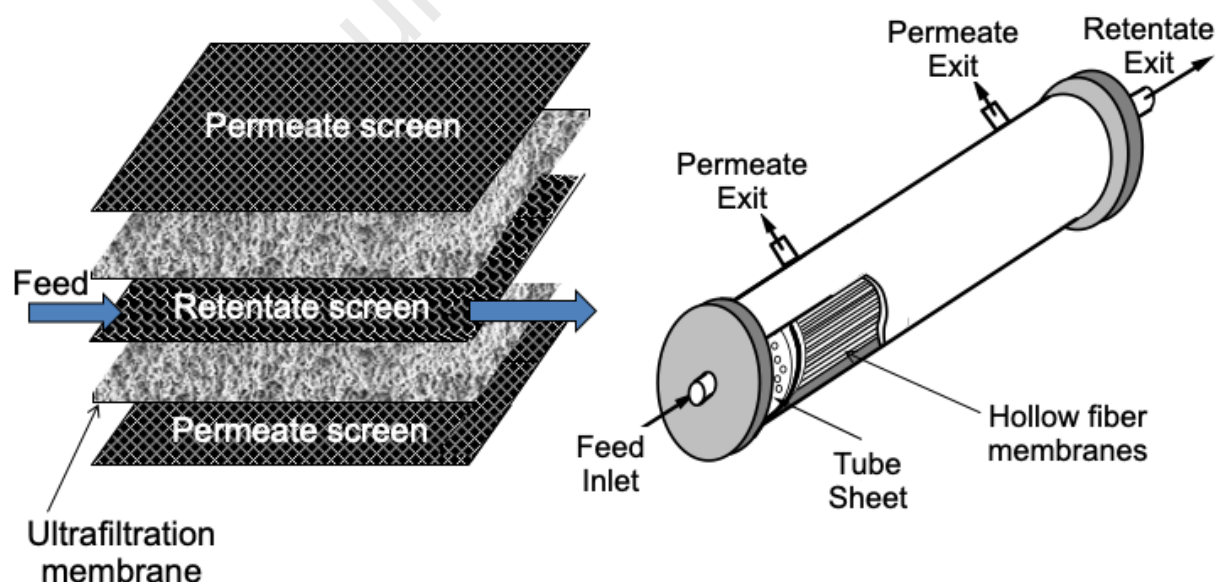


Figure 7 – Schematic of TFF cassette (left panel) and hollow fiber module (right panel).

In addition to conventional TFF modules that have been adapted to single-use format, it is also possible to develop ultrafiltration modules specifically targeted to single-use applications. For example, Yehl and Zydney [38] explored the use of high-flux polysulfone hollow fiber hemodialysis cartridges operated at very low filtrate flux (below 0.3 $\mu\text{m/s}$ or 1 LMH) for protein concentration in a single pass TFF mode; the very low flux significantly reduced the extent of membrane fouling, leading to stable operation with continuous protein concentration over more than 5 days without any physical or chemical cleaning. This is particularly attractive for continuous bioprocessing as discussed later in this Review. These hemodialysis modules cost only \$15 / m^2 (or less), more than an order of magnitude less than conventional UF modules for bioprocessing. In addition, these modules are provided in a pre-sterilized format with aseptic connectors and tubing assemblies, originally designed for direct application in an extracorporeal blood circuit. This system can thus provide single pass TFF performance but at a price that is compatible with disposable (single use) technologies.

The flux – pressure profiles for these modules (Figure 8) look very similar to traditional UF modules, with the flux initially increasing with increasing transmembrane pressure (TMP) but then approaching a nearly pressure-independent value at high TMP when the flux is limited by the concentration polarization boundary layer (Equation 1). The solid curve is a model calculation developed by Yehl and Zydney [38] accounting for the effects of the concentration-dependent viscosity at high IgG concentrations:

$$TMP = \frac{q_p}{L_p} \left[A - \frac{q_{feed}}{k_o} \int_{x_{out}}^1 \frac{\exp\left(\frac{[\eta]C_{feed}}{(x-\frac{K}{S}[\eta]C_{feed})}\right) dx}{\ln\left(\frac{x C_w}{C_{feed}}\right)} \right]^{-1} \quad (3)$$

where q_p and q_{feed} are the permeate and feed flow rates, L_p is the membrane permeability, k_o is the mass transfer coefficient in the limit of an infinitely dilute solution, $x = q/q_{feed}$ is the ratio of the local retentate flow rate to the inlet feed flow rate, $[\eta]$ is the intrinsic viscosity, S is the shape factor or Simha parameter, and K is the crowding factor. The solid curve in Figure 8 is developed using $[\eta] = 0.0086$ L/g, $K/S = 0.19$, and $C_w = 400$ g/L.

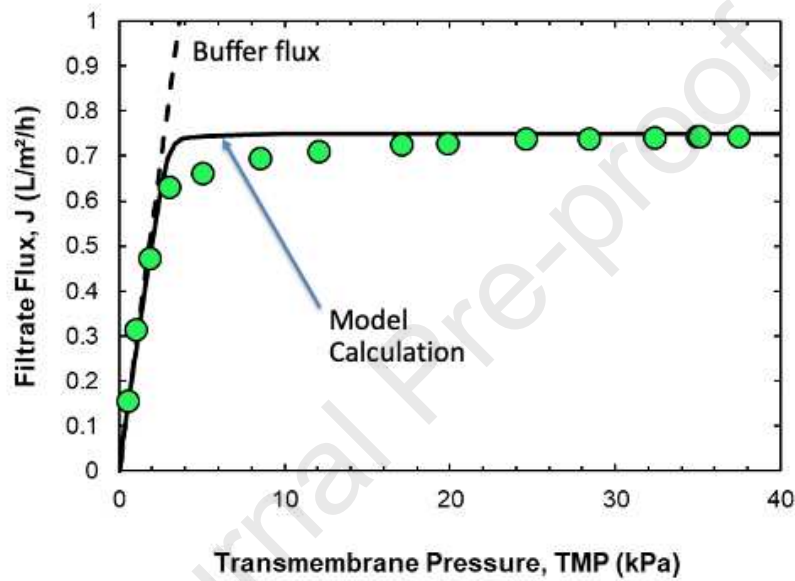


Figure 8: Filtrate flux as a function of transmembrane pressure for a hollow fiber module operated with $q_{feed} = 25$ mL/min (corresponding to a feed flux of 0.8 LMH) and $C_{feed} = 15$ g/L. The solid curve is the model calculation given by Equation (3); the dashed line shows the membrane limited flux. Reproduced from Yehl and Zydney [38] with permission.

3. Continuous Bioprocessing

Current bioprocessing operations are performed largely in batch mode but concerns about biomanufacturing costs and the need for greater manufacturing flexibility has led to growing interest in the potential of continuous bioprocessing, either for specific (linked) steps or for the

entire process, with the product moving from one step to the next without any significant intervening hold tanks. In principle, continuous bioprocesses can provide [10]:

- significant reductions in facility size and capital equipment costs
- significant increases in productivity (since all equipment is in use at all times)
- greater flexibility (due to more efficient utilization of the facility)
- improved product quality due to steady-state operation

The potential to improve product quality is of particular interest in the production of complex biological products. Continuous processes have shorter and more uniform residence time distribution than batch systems, leading to reductions in the variability in the extent of glycosylation [39], deamidation [40], and the level of protein aggregation, e.g., due to unfolding interactions on the surface of membranes or chromatographic resins [41] – [43]. Janet Woodcock, Director of the Center for Drug Evaluation and Research (CDER), specifically identified continuous manufacturing as an important step in the modernization of pharmaceutical manufacturing [44].

In addition, there is growing pressure on biomanufacturers to reduce the price of biopharmaceuticals, both to combat the growing overall cost of healthcare and to increase the availability of novel therapeutics in developing countries. Moreover, lower cost manufacturers are providing increased competition for established biopharmaceutical companies, particularly through the production of generic versions of biological molecules (commonly referred to as biosimilars). Walthe et al. [45] estimated that an integrated continuous biomanufacturing process would have 55% lower costs than a conventional batch process for the production of a monoclonal antibody product. Hammerschmidt et al. [46] reported even larger reductions in both operating costs (by nearly a factor of 5) and capital costs (by more than a factor of 3) for non-

antibody products. Continuous processes can also enhance the flexibility of manufacturing facilities designed to produce multiple products, enabling companies to more effectively manage rapid changes in product portfolios and demand.

There have been several recent Reviews of technologies for continuous biomanufacturing, including papers by Zydney [10], Fisher et al. [47], Schofield [48], and Rathore et al. [49] among others. The current discussion is focused specifically on the role membrane technology can play in the development of successful perfusion bioreactors, in the purification of recombinant proteins (in combination with precipitation and adsorption), and in continuous product formulation.

3.1 TFF / ATF for Perfusion Bioreactors

Continuous cell culture technology using perfusion bioreactors was developed more than three decades ago [50]. Perfusion bioreactors operate with continuous addition of nutrients coupled with continuous removal of product (and soluble impurities) along with periodic purging of cells / debris to maintain a nearly constant bioreactor volume and cell density. The short residence time in these systems is particularly attractive for the production of highly labile products that are susceptible to degradation, aggregation, and / or proteolysis within the bioreactor [51]. Perfusion systems can have cell densities nearly 10-fold greater than typical batch bioreactors, leading to enhanced productivity. The use of perfusion bioreactors for monoclonal antibody (mAb) production is now well-established [52] – [54].

The need to recycle the retained cells back to the perfusion bioreactor places significant constraints on the clarification step since maintaining high cell viability is needed for long-term operation of the bioreactor. Thus, centrifugation and depth filtration, which are the primary

methods for initial clarification in batch processes, are rarely used for continuous bioreactors. Instead, cell recycle and clarification are typically performed using tangential flow filtration (TFF) with relatively large ID (inner diameter) hollow fiber membranes. The retentate stream, which contains the retained cells, is returned directly to the bioreactor while the permeate stream, which contains the secreted product, is taken for subsequent purification. Operation is at constant filtrate flux, with the pressure rising as the membrane fouls.

Much of the recent effort has focused on the development of membranes / processes that reduce the extent of fouling and minimize product retention, allowing a single module to be used for multiple weeks without cleaning or replacement. Although antibody transmission through the microfiltration membranes is 100% at the start of the process, numerous studies have documented a significant increase in product retention over the course of operation (Figure 9). Wang et al. [55] demonstrated that this “product sieving” is due to fouling associated with cell debris that is approximately 100 nm in size (just smaller than the 200 nm pore size of the membrane). It was possible to significantly reduce the generation of this debris by using a low shear centrifugal pump (from Levitronix), in contrast to the peristaltic pumps used in traditional TFF systems. There is also evidence of membrane fouling by micelles formed by the antifoam agents used to minimize foaming in the bioreactor [56].

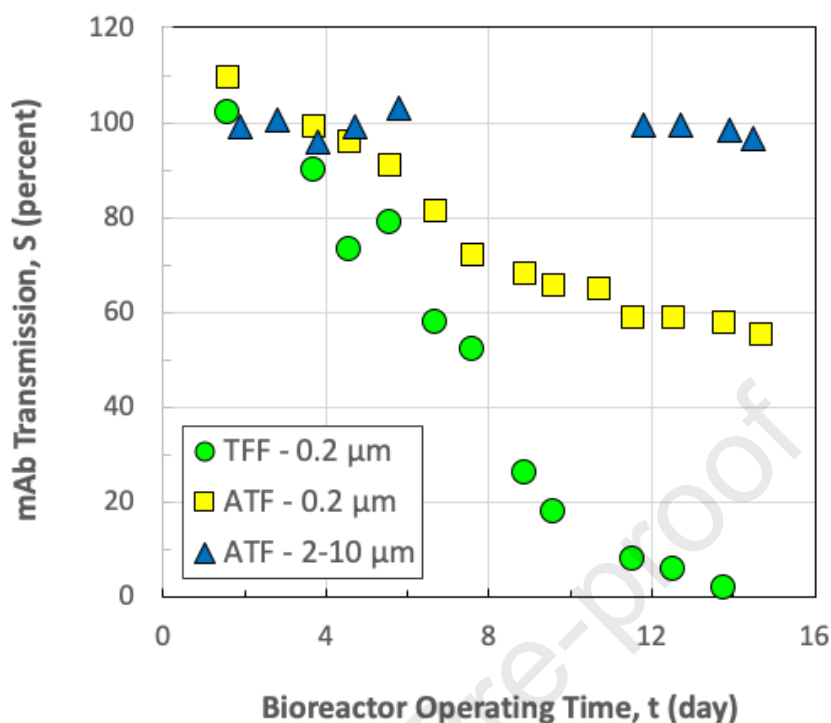


Figure 9: Antibody transmission (ratio of mAb concentration in filtrate to that in bioreactor) during operation of a CHO cell perfusion bioreactor using TFF and ATF modules with 0.2 μm polysulfone [55] and 2-10 μm polyethylene [57] hollow fiber membranes.

Pinto et al. [58] evaluated the effects of membrane polymer and pore size on fouling and product retention during operation of a TFF system for a CHO cell perfusion bioreactor producing a monoclonal antibody product. Similar performance was obtained with several 0.2 μm pore size membranes, with slightly less product retention for the polysulfone, polyethersulfone, and modified polyethersulfone compared to that observed with a mixed cellulose ester. Significantly less fouling was obtained using a macroporous (1-4 μm pore size) polyethylene membrane, with stable operation (nearly 100% product transmission) maintained over more than 3 weeks. However, these macroporous membranes gave a more turbid permeate that needed to be treated by depth filtration to avoid clogging of the subsequent chromatography

columns. Although most TFF systems for perfusion applications use hollow fiber membranes, MilliporeSigma has developed the Cellicon™ Perfusion Filter which uses a 5 µm pore size flat sheet Durapore® polyvinylidene fluoride membranes [59]. Limited data obtained with a CHO cell line show capacities of 8000 L/m².

A number of studies have shown that it is possible to significantly improve the performance of TFF processes used with perfusion bioreactors by employing what is known as Alternating Tangential Flow filtration (ATF). ATF typically uses large inner diameter hollow fiber membranes, but the flow is pumped through the module in an alternating fashion by a diaphragm pump that withdraws and returns the cells to the bioreactor in a cyclical fashion, with typical cycle times on the order of 1 min [56, 60]. This minimizes cell damage that can occur with higher shear peristaltic pumps, and the alternating flow direction can “un-clog” fibers that might become blocked by cells or cell debris at the fiber inlet. Filtration occurs in both directions, i.e., during both the withdrawal (pressure) and return (exhaust) cycles (Figure 10). In addition, the ATF systems have only a single connection to the bioreactor, which helps maintain sterility and facilitates replacement of the module. ATF has been very successfully employed during cell expansion (in the N-2 and N-1 bioreactors where N is the bioreactor used for commercial production).

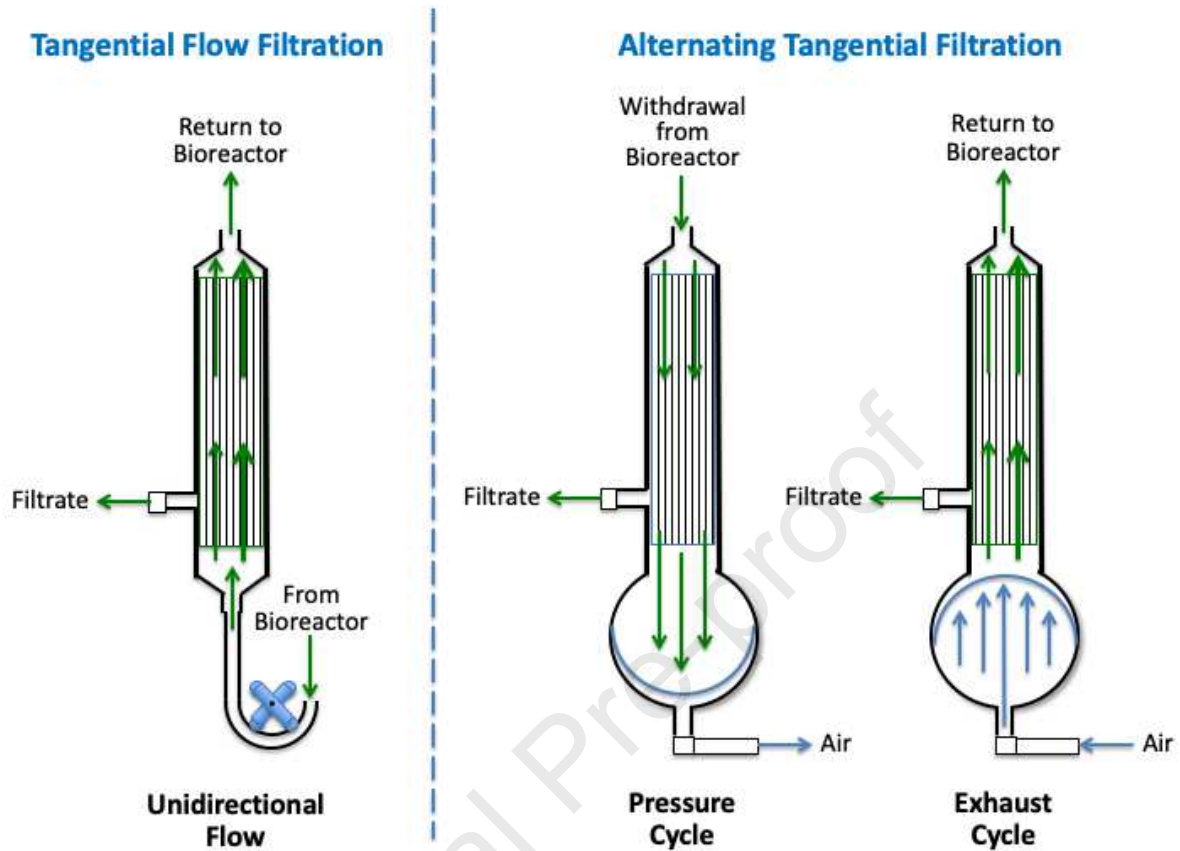


Figure 10: Schematic comparing the operation of tangential flow filtration (TFF) with alternating tangential filtration (ATF).

The ATF system is typically operated at very low filtrate flux, on the order of $0.3 - 2 \mu\text{m/s}$ (1-5 LMH), using feed flux (feed flow rate normalized by the membrane area) around 300 – 800 LMH. The combination of low flux, high feed flow rates (typical shear rates of $1,000 - 5,000 \text{ s}^{-1}$), and high membrane permeability generates a Starling recirculation flow in the module, with the large axial pressure drop leading to a reverse flow near the retentate exit [61]. This backflushing can clean much of the membrane length since the alternating flow shifts the location of the backflushing from one end of the module to the other.

Several studies have demonstrated that ATF operation has significantly less fouling and product retention than corresponding TFF systems [60, 62], with typical results summarized in Figure 8. ATF systems integrated directly with perfusion bioreactors have been operated continuously for several weeks. Walther et al. [63] showed that long residence times in the ATF module led to decreased cell growth rates and productivity, while high hydrodynamic stresses led to reduced cell size and specific productivity. Membrane fouling became significant at filtrate flux above $0.7 \mu\text{m/s}$ (2.5 LMH). The performance can be further improved using larger pore size membranes; Wang et al. [57] showed stable operation with no measurable antibody retention using $5 \mu\text{m}$ pore size alpha alumina membranes (Pall) and 2-10 μm pore size hollow fiber polyethylene membranes (Spectrum, now part of Repligen) as shown in Figure 9.

3.2 Membrane-enhanced Separations

Although a number of multi-column chromatographic processes have been developed for continuous bioprocessing, these systems typically work cyclically with large variations in product concentration (and buffer conditions) as the columns move through the binding, washing, and elution steps [7]. In addition, column chromatography becomes less attractive as the product titer increases since the volume of the expensive resin must be increased to accommodate the greater product mass. There have thus been a number of efforts focused on the development of truly continuous capture steps that can be connected directly to the outflow from a perfusion bioreactor as part of a truly integrated continuous biomanufacturing process. Membrane technology can play a critical role in this application, with the actual separation / purification of the protein product accomplished by selective precipitation or continuous countercurrent tangential chromatography.

3.2.1 Continuous Precipitation

Precipitation was one of the first techniques used for protein purification. Plasma-derived proteins are still purified by precipitation using the Cohn fractionation process, using a combination of alcohol, salt, pH, and temperature [64]. Several recent studies have demonstrated that it is possible to purify monoclonal antibody products by precipitation from clarified cell culture fluid using a variety of salts and polymers. Although precipitation was traditionally accomplished in batch mode, continuous tubular precipitation reactors can provide more uniform residence time and precipitate morphology. Host cell proteins and DNA can then be removed by washing with an appropriate buffer, with the precipitated protein recovered by tangential flow microfiltration while the soluble impurities are removed in the permeate [65]. mAbs have been successfully precipitated using a variety of precipitants and flocculants, including ethanol, various salts (CaCl_2 and ZnCl_2), polymers (e.g. polyethylene glycol), and caprylic acid, both alone and in various combinations.

Li et al. [66] showed that high degrees of purification can be achieved using countercurrent staged microfiltration modules as shown in Figure 10. In this case, mAb precipitation was driven by a combination of a crosslinking agent (ZnCl_2) and a volume exclusion agent (polyethylene glycol = PEG). The precipitate (approximately 2% solids) was then dewatered by microfiltration before washing. The extent of impurity removal (R) in countercurrent staged washing process is given as [66]:

$$R = \frac{C_{feed}}{C} = \frac{\alpha^{N+1}-1}{\alpha-1} \quad (4)$$

with

$$\alpha = \frac{Q_{wash}}{Q_{feed}} = \frac{x}{1-x} \quad (5)$$

where C_{feed} is the concentration of soluble impurities in the feed entering the wash step, C is the concentration of soluble impurities leaving the wash step, Q_{feed} is the feed flow rate, Q_{wash} is the flow rate of the wash buffer, N is the number of stages, and x is the conversion in the membrane module (equal to the ratio of the permeate flow rate to inlet feed flow rate for the membrane module). A 2-stage wash step using $\alpha = 9.5$ can provide more than a 100-fold reduction in the concentration of soluble (non-retained) impurities.

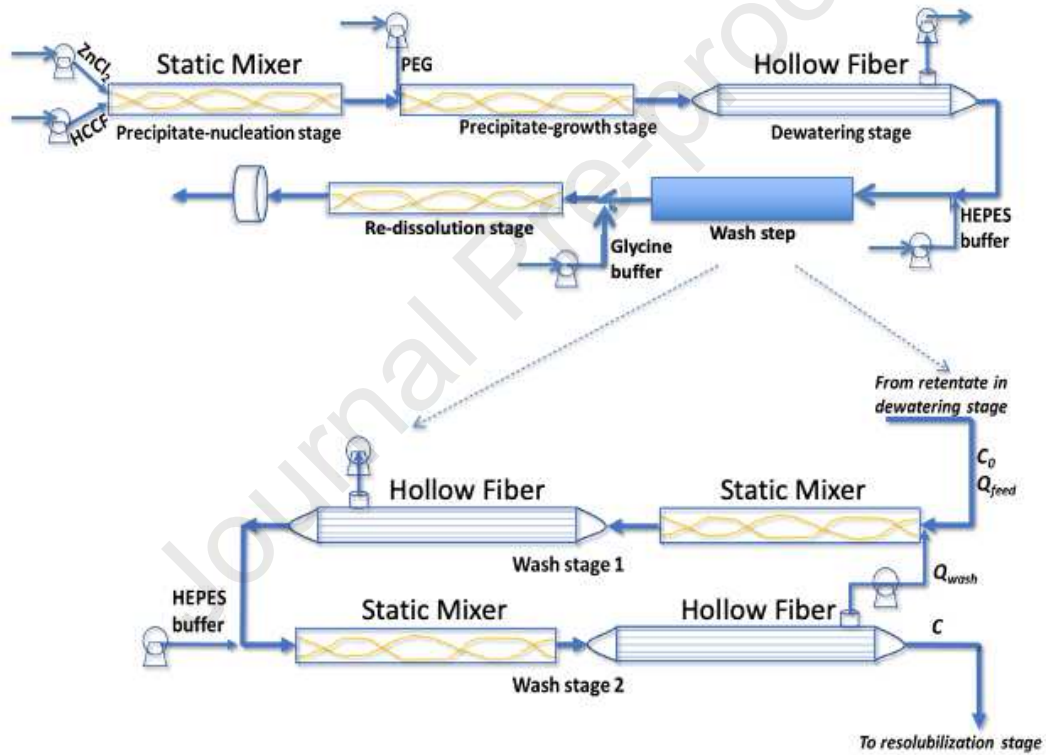


Figure 11: Schematic of a continuous precipitation system incorporating a 2-stage countercurrent washing step.

Successful application of this type of continuous precipitation process requires that the microfiltration modules can be used for extended periods of time without fouling. This can be

achieved by operating the microfilter at a filtrate flux below the critical filtrate flux, defined as the maximum value of the filtrate flux below which fouling is negligible [67]. Typical data for the critical flux for precipitated human serum IgG are shown in Figure 12 [68]. The straight lines represent the shear rate dependence given by the classical concentration polarization model using the Brownian diffusivity ($\gamma^{1/3}$), the concentration polarization model using the shear-induced diffusivity (γ^1), and the inertial lift model (γ^2) as discussed by Belfort et al. [69]. The data for the IgG precipitate seem to lie between the polarization models using the Brownian and shear-induced diffusion models.

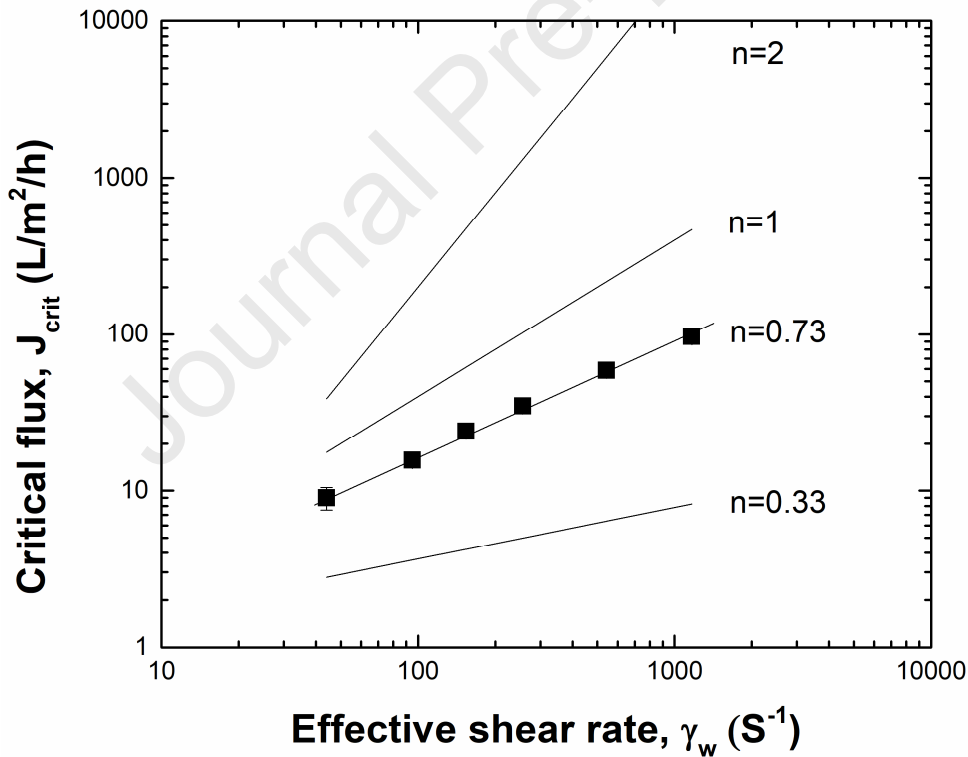


Figure 12: Critical flux (as a function of the effective wall shear rate for data obtained during microfiltration of IgG precipitates formed using 5 g/L IgG, 10 mM ZnCl₂ and 7 weight percent PEG. Solid lines represent shear rate dependence (slope) of $n=0.33$ for Brownian diffusion model, $n=1$ for shear induced diffusion model, and $n=2$ for inertial lift model. Also shown is linear regression fit to the data on a log-log plot (slope = 0.73). From [68].

3.2.2 Continuous Countercurrent Tangential Chromatography

Continuous countercurrent tangential chromatography (CCTC) uses ion exchange, affinity, or mixed mode resins particles in the form of a flowing slurry, with the solid-liquid separations accomplished by hollow fiber membrane modules as shown in Figure 13 [70]. For example, when used in a bind-and-elute format, the product binds to the resin with soluble impurities (host cell proteins and DNA) removed through the membrane during the washing stage. The product is then collected in the permeate by adding an appropriate elution buffer to reverse the binding interactions. Each step in the chromatographic process (binding, washing, elution, stripping, and regeneration) uses a combination of hollow fiber membrane modules and static mixers, with the mixers providing the required residence time for binding of the protein to the resin. The hollow fiber membrane modules are arranged in a countercurrent configuration, analogous to that shown in the inset in Figure 11, to maximize impurity removal and minimize buffer usage.

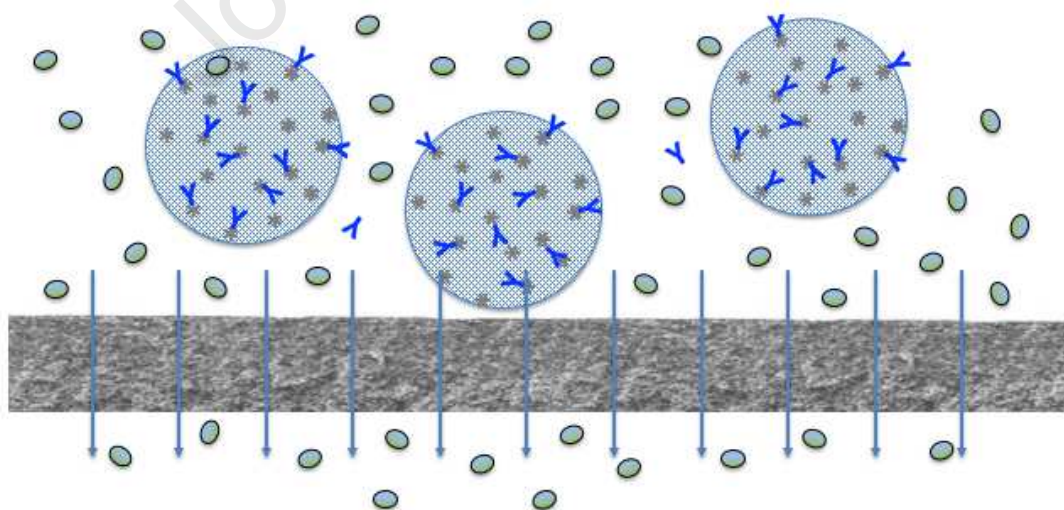


Figure 13: Schematic of continuous countercurrent tangential chromatography (CCTC) for impurity removal (filled ovals) from a bound monoclonal antibody (Y-shaped).

The CCTC system has a number of distinct advantages compared to traditional column chromatography. First, the CCTC system is operated continuously and at true steady state, which is ideal for incorporation in an integrated continuous bioprocess [71]. Second, since the resin is used as a flowing slurry, the pressure drop in CCTC is typically < 15 psi, enabling the flow path to be made of completely disposable plastic materials, in contrast to the stainless steel columns used in commercial-scale chromatography. Thus, CCTC can be effectively employed in the development of single-use biomanufacturing facilities, enhancing flexibility and reducing capital costs. CCTC have shown productivities as high as $100 \text{ g mAb / L of resin / hr}$, which is several times that of conventional packed columns. This high productivity is a direct result of having all of the resin fully utilized at all times. CCTC systems are typically operated with binding capacities around 70-80% of the static (equilibrium) binding capacity for the particular resin.

3.3 Single Pass TFF

Typical ultrafiltration modules operate at relatively low conversion, defined as the ratio of the permeate to feed (inlet) flow rate. Thus, the required degree of concentration is achieved by continually recycling the retentate through the module while the permeate is removed. However, this is an inherently batch process, with the desired target product concentration not attained until the very end of the process.

A number of companies have developed ultrafiltration modules specifically for continuous bioprocessing, with the desired concentration factor (or volume reduction) obtained in a single pass through the module without any recycle; this type of process is typically referred to as Single Pass Tangential Flow Filtration or SPTFF. SPTFF has been used for in-line

concentration of the feed stream between unit operations, e.g., to reduce the process volume to eliminate tankage constraints [72, 73] or to increase the productivity of subsequent chromatography or precipitation steps [74]. For example, Elich et al. [75] obtained a four-fold increase in mAb loading (at the same HCP clearance) in an anion exchange column by preconcentrating the mAb feed using SPTFF. Arunkumar et al. [76] demonstrated the effectiveness of SPTFF as an inline concentration step for cell culture harvest obtained from a perfusion bioreactor. SPTFF can also be used in the final formulation of biotherapeutics, with the single pass operation enabling higher product recoveries, particularly for very high concentration antibody formulations [77]. In addition, the absence of any recycle stream can be attractive for concentrating products that are prone to aggregation or denaturation.

The first specially designed SPTFF module was developed by Gaston de los Reyes and Leon Mir [78] at SPF Innovations (now exclusively licensed to Pall and sold under the tradename CadenceTM). The Pall CadenceTM InLine Concentrator (ILC) consists of multiple internal stages with the number of parallel flow channels in each stage decreasing sequentially along the length of the module (Figure 14). This reduction in cross-sectional area for flow balances the reduction in the retentate flow rate that occurs as permeate is removed, resulting in the high conversions required for most SPTFF applications. The Pall CadenceTM SPTFF module has been used to concentrate serum IgG from 45 g/L to more than 200 g/L with greater than 95% product recovery [72] and for the in-line concentration of multiple monoclonal antibodies between various chromatography steps [73].

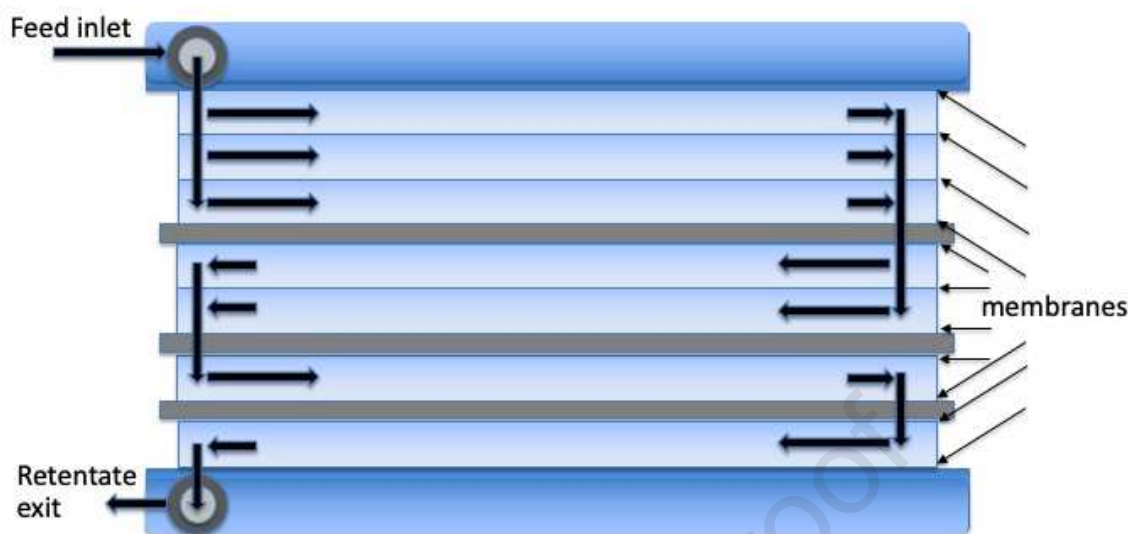


Figure 14: Schematic diagram of flow path in a multi-cassette single-pass tangential flow filtration (SPTFF) module with sequentially decreasing number of parallel flow channels (from 3 to 2 to 1 to 1).

It is also possible to achieve high conversions in a single pass format using more conventional UF modules by increasing the channel length and / or reducing the feed flow rate. The Pellicon® SPTFF system (MilliporeSigma) uses a stainless steel holder with appropriate diverter plates between a number of modules in series to obtain the desired flow path. In contrast, Yehl and Zydney [38] demonstrated that commercial high-flux hemodialysis cartridges could be used for SPTFF by operating the modules (with only a 25 cm long flow path) at very low feed flow rates. These modules successfully concentrated an Immunoglobulin G product from 15 g/L to more than 200 g/L in a single pass, with stable operation for more than 5 continuous days of operation without any cleaning or backflushing. In addition, these modules were sufficiently inexpensive that they could provide a disposable technology for single-use manufacturing facilities.

3.4 Continuous Diafiltration

Diafiltration is used for desalting and buffer exchange, both in conditioning feed streams prior to other downstream processes and as part of the final formulation to place the product in an appropriate solution for storage and delivery to the patient [7]. Diafiltration is traditionally accomplished using a constant volume batch system, with high levels of buffer exchange requiring between 5 and 10 diavolumes (where a diavolume is defined as the ratio of the total cumulative filtrate volume to the constant batch volume). This typically requires several hours of operation and close to 100 pump passes, with the diafiltered product not attained until the end of the process. Goodrich et al. [79] developed a tank cycling system that provides continuous inflow of feed and outflow of formulated / diafiltered product, but with batch operation of the individual tanks. This system has the advantage of using conventional UF modules and DF protocols, although the large residence time and multiple pump passes are not ideal for continuous processes.

An alternative approach for continuous buffer exchange is to use simple in-line dilution of the feed followed by continuous concentration by SPTFF. However, a one thousand-fold reduction in the concentration of a particular salt or impurity would require a ratio of dilution buffer to feed flow rate of 999, which is completely impractical. Rucker-Pezzini et al. [80] demonstrated that a series of sequential dilution and concentration steps can be used for continuous DF, with each concentration performed using a CadenceTM ILC. Data obtained using a 3-stage system showed 99.75% impurity removal, although the process had significantly greater buffer requirements than a traditional batch DF. Pall Corporation has developed an integrated Inline Diafiltration (ILDF) module which uses six repeated stages of dilution / concentration, each using a dilution buffer flow rate that is approximately 2.2x the feed flow.

This system can achieve an overall impurity removal of 99.9% with only slightly greater buffer consumption than a batch process.

More effective operation of the continuous diafiltration process can be achieved by arranging the SPTFF modules in a countercurrent configuration to increase the effectiveness of the staging [81, 82]. In this case, the feed is pumped into the final module in the cascade, while the diafiltration buffer is added to the first stage, with the retentate and permeate from each stage recycled in opposite direction (Figure 15). The fractional removal of a small impurity in this type of countercurrent staged system can be evaluated as:

$$f = \frac{1 - \beta}{1 - \beta^{N+1}} \quad (6)$$

where β is the ratio of the diafiltration buffer flow rate to the feed flow rate. Jabra et al. [82] showed that a 3-stage countercurrent diafiltration system with $\beta = 10$ provided more than 99.9% removal ($f < 0.001$) of a small impurity from an Immunoglobulin G product.

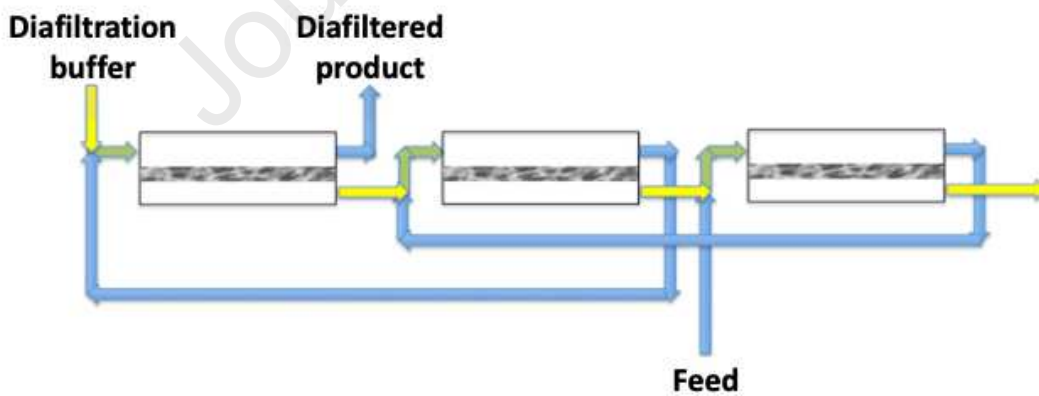


Figure 15: Three-stage countercurrent staged diafiltration. Feed enters in last module (on right) while diafiltration buffer enters in first module (on left).

An alternative approach for continuous buffer exchange is to use countercurrent dialysis, with small impurities removed (and fresh buffer added) by diffusion across the semipermeable membrane as shown in Figure 16 [83]. In this case, the degree of impurity removal is given as:

$$R = \frac{\alpha \exp(\beta) - 1}{\alpha - 1} \quad (7)$$

where

$$\beta = \frac{k_o A}{q_F} \left(1 - \frac{1}{\alpha} \right) \quad (8)$$

with $\alpha = q_D/q_F$, k_o equal to the overall diffusive mass transfer coefficient, and A the membrane area. Yehl and Zydney [83] showed that commercial hollow fiber dialyzers could provide more than 99.9% buffer exchange in single-pass operation using less than 0.02 L of buffer per g protein, which is significantly less than that used in current batch diafiltration. In addition, the performance of the countercurrent dialysis was largely independent of the protein concentration, in contrast to the significant reduction in filtrate flux at high protein concentrations seen in pressure-driven diafiltration processes. Moreover, the absence of any pressure-driven filtration significantly reduced the rate of membrane fouling, potentially providing a platform for long-term continuous buffer exchange without the need for any membrane cleaning.

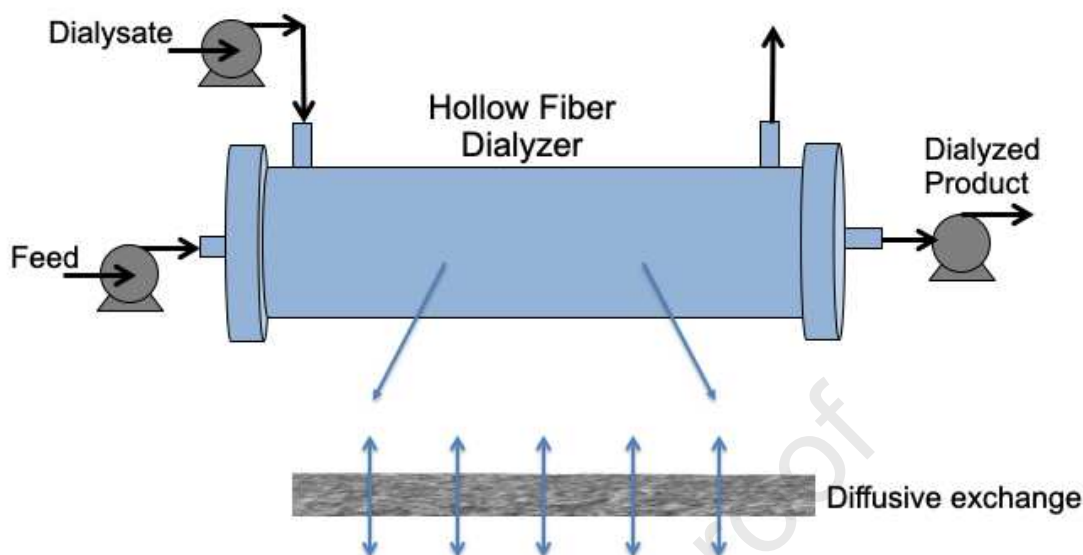


Figure 16: Countercurrent hollow fiber dialyzer for continuous buffer exchange with lower panel showing diffusive exchange across the semipermeable membrane.

More recently, Tan and Franzreb [84] developed a 3D-printed diafiltration module which enabled continuous addition and removal of filtrate through two parallel membranes that defined the feed flow path (Figure 17). This configuration is very similar to that described in the patent by Leuthold et al. [85]. The upper membrane was used for addition of the diafiltration buffer, while the lower membrane was used for removal of filtrate. The performance of a small prototype module with 5-cm flow length was examined using bovine serum albumin (BSA) as a model protein. The system was able to provide simultaneous concentration (by a factor of 4.6) and desalting (by a factor of two), although future efforts will be required to demonstrate the scalability and economic viability of this technology.

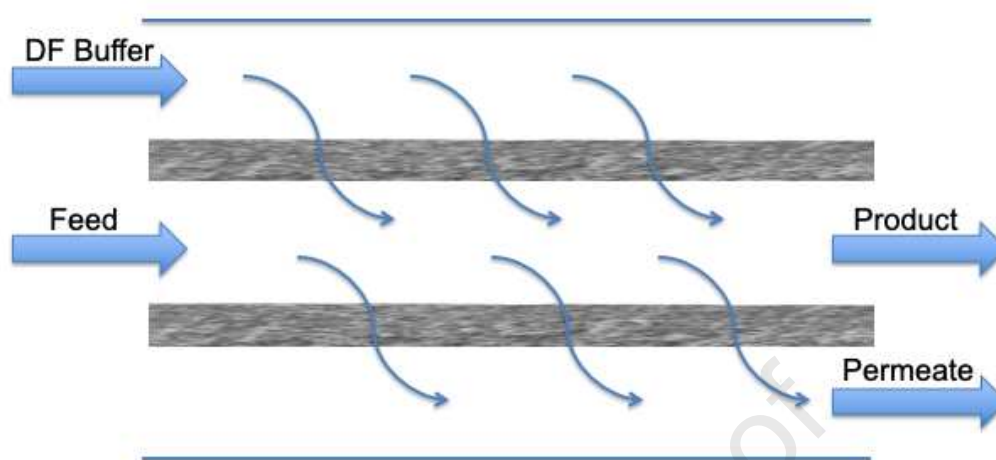


Figure 17: Countercurrent hollow fiber dialyzer for continuous buffer exchange.

4. Novel Biotherapeutics

Much of the growth in the biotechnology industry over the past two decades has been due to the success of monoclonal antibodies (mAbs) and mAb-related products, e.g., Fc-fusion proteins, Fab fragments, and antibody-drug conjugates (ADC) [86]. However, there is growing interest in the commercialization of more complex therapeutics, including live attenuated virus vaccines, viral vectors for delivery of gene therapy agents, plasmid DNA, glycoconjugate vaccines, and cell-based therapies. Although each of these biotherapeutics has its own manufacturing challenges, they are all considerably larger than recombinant proteins. The large size leads to significant diffusional mass transfer limitations in the development of chromatographic separations using traditional porous resins, and it also complicates the development of sterilizing grade filtration processes in which the product is collected in the permeate through a 0.2 μm pore size sterilizing grade filter. The use of membrane adsorbers as an alternative to column chromatography has already been discussed in Section 2.2; the

convective flow through these membranes is particularly attractive for large biotherapeutics. The discussion in the following section examines potential opportunities for the development of pressure-driven membrane processes for the downstream processing of glycoconjugate vaccines, plasmid DNA, viruses / viral vectors, and cell-based therapeutics.

4.1 Glycoconjugate Vaccines

Glycoconjugate vaccines are formed by covalent coupling (conjugation) of the outer capsular polysaccharide from an infectious micro-organism to a highly immunogenetic protein such as the tetanus toxoid or the non-toxic mutant of the diphtheria toxin [87]. These glycoconjugates provide a much stronger immune response than the polysaccharides alone, particularly in infants and the elderly. Glycoconjugate vaccines have been developed for immunization against *Haemophilus influenzae* type b (Hib), *Streptococcus pneumoniae*, *Salmonella typhi*, and *Neisseria meningitidis* [87].

One of the critical steps in the purification of these vaccines is the separation of the glycoconjugate from any residual protein and polysaccharide that remain after the conjugation reaction [88]. Although a number of chromatographic processes have been explored, including hydrophobic interaction, reverse phase, and size exclusion, these systems have relatively low throughput (due to mass transfer limitations) and poor selectivity (due to the similar chemical properties of the free polysaccharide and the glycoconjugate product). There has thus been considerable interest in the development of a selective ultrafiltration process that can effectively exploit the difference in size between the large glycoconjugate and the smaller protein / polysaccharide components.

The idea of using pressure-driven membrane processes for highly selective separations dates back to the early work on ultrafiltration [89]. van Reis et al. [90] discussed the principles underlying high performance tangential flow filtration (HPTFF) for protein purification, with follow-up studies demonstrating the use of this technology for the purification of a positively-charged monoclonal antibody fragment (Fab'2) produced in *E. coli* [91]. Most of these studies used prototype charged membranes, made by chemical modification of a base cellulose membrane with either sulfonic acid or quaternary amine groups. These charged UF membranes are not yet available commercially. HPTFF has not yet had a significant impact on commercial processing of mAbs or antibody-related products due to the challenges in displacing well-established chromatographic processes for antibody purification (and in some cases to the need for specially designed charged membranes to obtain the desired selectivity, although HPTFF technology does not require the use of charged membranes).

Emami et al. [92] examined the use of TFF for the purification of a glycoconjugate formed from a polysaccharide from the *Streptococcus pneumoniae* bacteria that contains a high concentration of glucuronic acid monomers giving it a significant negative charge. Typical data for the selectivity between the glycoconjugate and free polysaccharide:

$$\psi = \frac{S_{o, polysaccharide}}{S_{o, glycoconjugate}} \quad (9)$$

are shown in Figure 18. The selectivity initially increases with increasing filtrate flux due to higher degree of concentration polarization but then declines at large flux due to membrane fouling. Operation of the Pellicon XL cassette at the optimal flux ($13 \mu\text{m/s} = 47 \text{ LMH}$) provided more than 100-fold purification of the glycoconjugate with 70% yield [92], which is highly competitive with chromatographic separations but with much greater throughput and lower overall cost. Additional studies will be required to demonstrate this technology for commercial

manufacturing of these glycoconjugate vaccines, including any issues associated with scale-up from the Pellicon XL cassette (with 50 cm² membrane area) to commercial TFF systems with multiple m² of membrane.

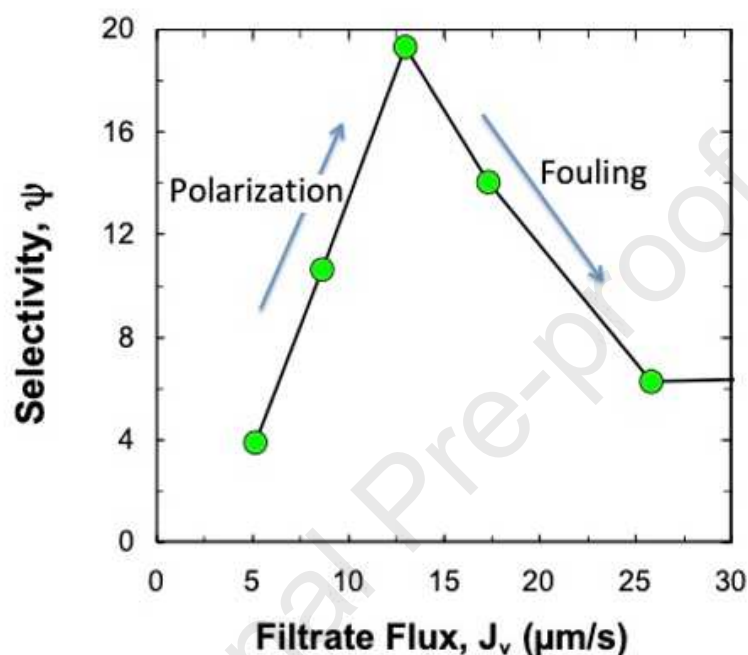


Figure 18: Selectivity as a function of the filtrate flux for the separation of a glycoconjugate vaccine from free polysaccharide using tangential flow filtration with Pellicon XL cassettes with composite regenerated cellulose membranes. Adapted from [92].

Another challenge in the downstream processing of glycoconjugate vaccines is the final (sterilizing grade) filtration step. The glycoconjugates are typically 100 – 300 nm in size (as determined by dynamic light scattering [93]), which is comparable to the 200 nm pore size of sterilizing grade filters [7]. This leads to significant fouling as shown by the rapid rise in transmembrane pressure during operation at constant filtrate flux and the relatively low

capacities of $<100 \text{ L/m}^2$ (Figure 19). The solid curve in Figure 19 is given by a modified form of the intermediate pore blockage model [93]:

$$\Delta P = (\Delta P_0 - \Delta P^*) \exp\left(\frac{\beta V}{A}\right) + \Delta P^* \quad (10)$$

where ΔP_0 is the initial pressure drop, V/A is the volumetric throughput (equal to the total collected filtrate volume, V , divided by the area of the membrane, A), and β is the pore blockage parameter. Equation (10) is developed based on the assumption that fouling occurs only over a limited portion of the thickness of the membrane, with ΔP^* equal to the pressure drop over the region of the membrane that remains unfouled. This assumption was verified using confocal microscopy to identify the location of fluorescently-labeled glycoconjugates within the fouled membrane, with confocal images showing glycoconjugate deposition only in the upper $5 \mu\text{m}$ depth of the filter.

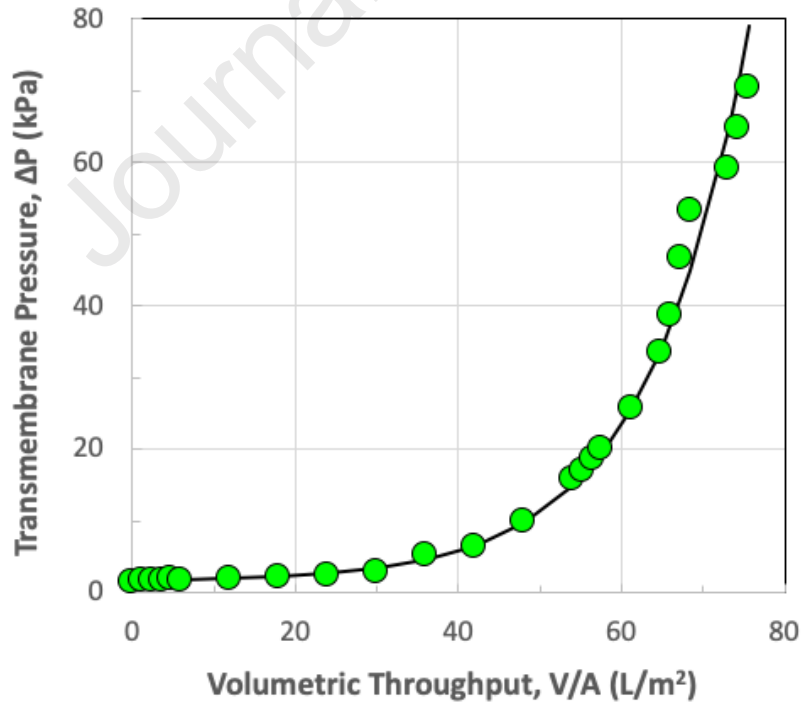


Figure 19: Fouling of a $0.22 \mu\text{m}$ Durapore® sterilizing grade filter during constant flux filtration of a glycoconjugate vaccine at a flux of $20 \mu\text{m/s}$ (72 LMH). Data from

[93]. Solid curve given by Equation (10) with $\beta = 81 \text{ m}^{-1}$, $\Delta P_0 = 1.7 \text{ kPa}$, and $\Delta P^* = 1.5 \text{ kPa}$.

4.2 Plasmid DNA

Gene therapy refers to the use of DNA (or RNA) for the replacement, inactivation, or modification of one or more genes [94]. The early focus of gene therapy was in the treatment of hereditary diseases characterized by a single missing or malfunctioning gene, e.g., the introduction of a gene that codes for a missing clotting factor (most commonly Factor VIII) in the treatment of hemophilia. However, it is also possible to use gene therapy for the treatment of cancer using genetically-engineered T cells (discussed subsequently), and both DNA and RNA can be used as vaccines. In the latter case, the nucleic acids provide for the *in situ* production of a target antigen which then stimulates the patient's immune system [95].

DNA can be delivered to the cells of interest using either viral or non-viral vectors; some of the issues associated with the purification of viral vectors is discussed in the next section. Non-viral vectors include a range of polymers, peptides, nanoparticles, and lipids [96]. The DNA itself is in the form of a plasmid, a small extrachromosomal DNA molecule. Plasmids can exist in one of three topological isoforms: supercoiled, linear, and open-circular (or nicked). The supercoiled isoform is generated when the plasmid is made through the enzymatic “winding” of the DNA; this is the form that has the greatest therapeutic activity. The open-circular isoform is generated by a breakage in one of the DNA strands (referred to as “nicked”) while the linear isoform is produced when both DNA strands are cut. The linear and open-circular isoforms are considered process impurities and must be reduced to less than 20% of the total DNA per FDA guidelines [97]. This is a particularly challenging separation given that the different DNA isoforms have identical numbers of base pairs and base sequence.

Membrane systems are particularly attractive for the purification of plasmid DNA. Manzano et al. [98] showed that TFF could be used for initial purification of plasmid DNA, with host cell proteins and other small impurities washed through the membrane and removed in the permeate. Latulippe et al. [99] demonstrated that transmission of large plasmids (radius of gyration between 50 and 200 nm) through relatively small pore size ultrafiltration membranes (pore radius between 5 and 15 nm) is due to elongation of the plasmid DNA in the converging flow-field approaching the membrane pores (Figure 20). At low filtrate flux, the DNA remains highly coiled and is unable to enter the pores. The DNA elongates at higher filtrate flux due to the increase in the elongational forces, resulting in a critical value of the filtrate flux that depends on the membrane pore size but is essentially independent of the DNA size (i.e., number of base pairs).

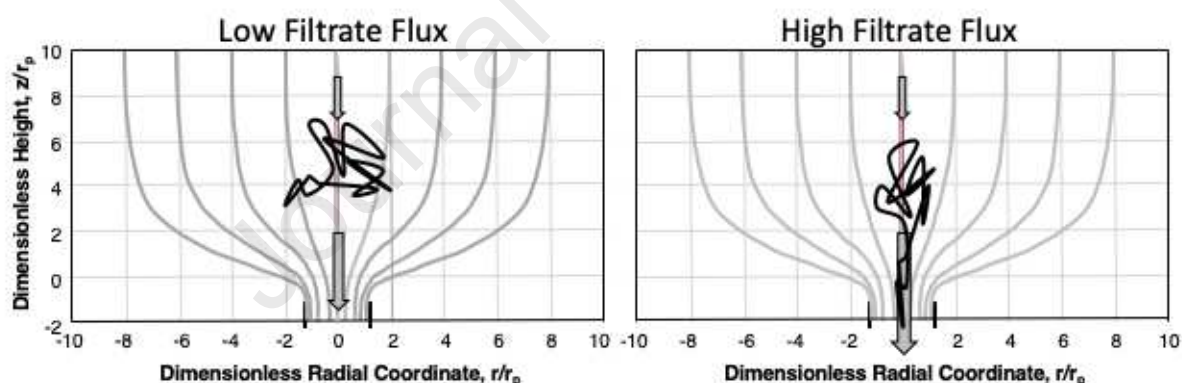


Figure 20: DNA elongation in the converging flow-field at the entrance to a membrane pore. Left panel – low filtrate flux. Right panel – high filtrate flux.

The critical filtrate flux for transmission is different for the supercoiled, open-circular, and linear isoforms due to differences in the elongational flexibility of these DNA molecules. The linear isoform has the lowest critical flux since it is most easily stretched and aligned with

the flow streamlines entering the membrane pore. The open-circular topology has the largest critical flux due to the bending “stiffness” of the expanded ring structure. Latulippe and Zydney [100] exploited these differences to develop membrane filtration processes for the purification of the desired supercoiled DNA. Typical results are shown in the agarose gel in Figure 21 for a 3.0 kbp (kilo base pair) DNA with an effective radius of 73 nm as determined by dynamic light scattering. All 3 isoforms are completely retained by the membrane at a flux of 10 $\mu\text{m/s}$ (36 LMH); these conditions would be appropriate for initial concentration of the DNA. Increasing the filtrate flux to 27 $\mu\text{m/s}$ (97 LMH) allowed the linear plasmid to be removed in the permeate while the supercoiled and open-circular isoform are recovered in the retentate. The supercoiled isoform could then be separated from the open-circular isoform by operating at a higher filtrate flux (between 55 and 85 $\mu\text{m/s}$), which is between the critical flux values for the two isoforms. It should be noted that these data were obtained in dilute solutions $\approx 1 \text{ mg/L}$; experiments at higher DNA concentrations tend to cause membrane fouling, although this can be reduced using backpulsing [101]. Li et al. [102] showed that further improvements in performance could be achieved using membranes with tapered pores to pre-stretch the DNA before the tight selective layer of the ultrafiltration membrane, opening up the opportunity to develop membranes with tailored pore geometries specifically designed for DNA purification.

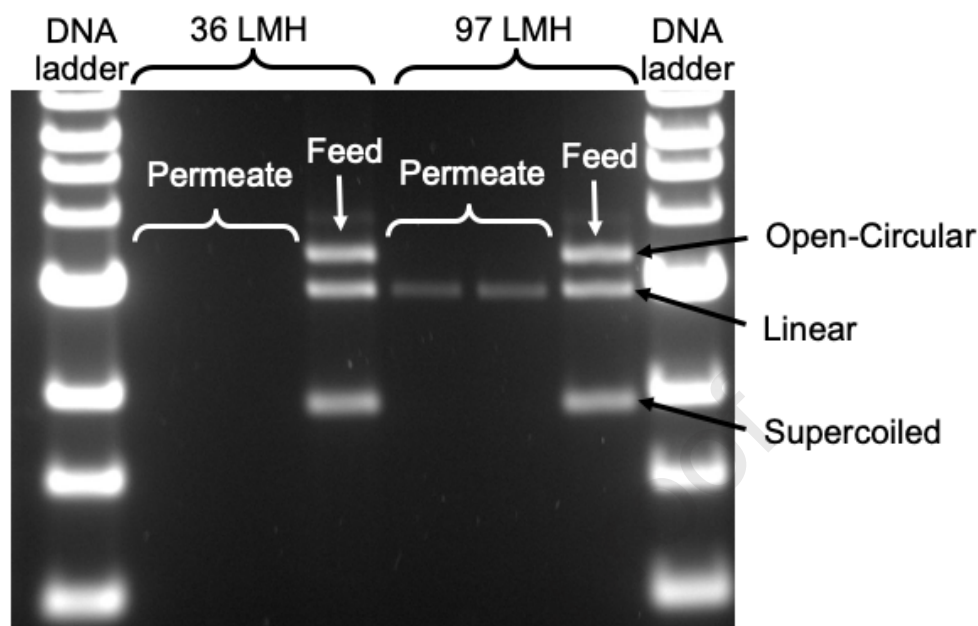


Figure 21: Agarose gel showing separation of a 3.0 kbp plasmid DNA by ultrafiltration (adapted from [100]). Lanes 1 and 8 are linear DNA ladders. Lanes 2 and 3 are filtrate samples collected at a filtrate flux of 36 LMH; Lane 4 is the final feed under these conditions. Lanes 5 and 6 are filtrate samples showing the linear isoform collected at a filtrate flux of 97 LMH; Lane 7 is the final feed under these conditions.

4.3 Viruses: Viral Vectors and Vaccines

One of the first applications of a live virus as a therapeutic was in the pioneering work by Edward Jenner on the development of the smallpox vaccine in 1796 [103], a century before the actual identification of an actual virus. Vaccines based on live-attenuated viruses (e.g., measles, mumps, and chickenpox), inactivated viruses (e.g., influenza and hepatitis A), and virus-like particles (e.g., human papilloma virus and hepatitis B) are currently used to provide vaccination against a wide range of infections [104]. The first oncolytic virus, a genetically-engineered form of the herpes simplex virus (Talimogene laherparepvec or T-VEC) that can kill melanoma tumor

cells, was approved for treating cancer in the U.S. in 2015 [105]. In addition, virus particles based on the adenovirus, lentivirus, and adeno-associated virus (AAV) are currently being explored as vectors for the delivery of gene therapy agents. AAV are particularly attractive for targeting neurological disorders (e.g., Parkinson's and Huntington disease) due to their small size and enhanced safety arising from the need for co-infection with a "helper" virus to achieve efficient replication [106]. Luxturna, the first FDA-approved gene therapy for treatment of inherited retinal disease, is delivered by an AAV.

Therapeutic viruses are produced in either an appropriate host organism (e.g., chicken eggs for measles and most influenza vaccines) or by cell culture (e.g., yeast for the hepatitis B vaccine or human embryonic kidney (HEK 293) cell lines for T-VEC). These viruses range in size from around 20 nm to as much as 1000 nm (Table 1), which can create significant challenges throughout the downstream purification process compared to the processing of much smaller proteins. Although the general strategy for the downstream process parallels that shown in Figure 1, additional steps are often required to release the virus from the host cells (by chemical lysis, homogenization, or sonication), and it is often necessary to remove incomplete and unfilled virus capsids as part of the purification process. In addition, very large viruses (like measles) cannot be sterile filtered.

Table 1 - Characteristics of typical therapeutic viruses

Virus	Clinical Application	Size	Production
Herpes simplex virus	Cancer (oncolytic)	155 – 240 nm	Vero (monkey) cells
Adenovirus	Cancer / gene therapy	≈90 nm	HEK 293
Adeno-associated virus	Gene therapy	≈20 nm	Insect cells / HEK 293
Lentivirus	Gene therapy	80 – 100 nm	HEK 293
Influenza virus	Vaccine	80 – 120 nm	Chicken embryo
Measles	Vaccine	≈1000 nm	Chicken embryo

As discussed previously, depth-filtration is very attractive for clarification of complex feed streams and has been used successfully with influenza, hepatitis, and rotavirus-like particles, all produced in insect cell lines [107]. The human Influenza A virus has also been clarified directly from the cell culture using both depth filtration and microfiltration [108]. Similarly, membrane adsorbers are very attractive for virus purification since the large size of the virus particles leads to significant diffusional mass transfer limitations in conventional packed column (bead) chromatography. Sulfated cellulose membrane adsorbers have been developed for purification of Influenza virus, while ion-exchange nanofiber membrane adsorbers have shown attractive performance with both Adenovirus type 5 [109] and lentiviral particles [110]. Kawka et al. [111] recently demonstrated the use of laterally-fed membrane chromatography (discussed previously) with Sartobind® Q membranes for purification of human Adenovirus type 5 using bind-and-elute operation.

Large molecular weight cutoff ultrafiltration membranes (MWCO > 100 kDa) can be used for concentration and buffer exchange as well as for removal of host cell proteins and other

impurities from larger viruses. For example, Carvalho et al. [112] demonstrated high removal of both DNA and HCP using 1000 kDa nominal molecular weight cutoff polyethersulfone membranes in Pellicon XL cassettes with nearly 100% recovery of an influenza virus-like particle (as determined by the hemagglutinin concentration). A tighter UF membrane (300 kDa) was needed for the final UFDF to minimize yield loss. Ultrafiltration of smaller adeno-associated viral vectors requires the use of tighter membranes. Arunkumar and Singh [113] found that high retention of AAV2 and AAV9 serotypes could only be achieved over a broad range of operating conditions when using 30 kDa composite regenerated cellulose membranes.

Sterilizing grade filtration is also highly challenging for viral products, particularly for some of the larger viruses and virus-like particles. Shoaebargh et al. [114] evaluated the performance of several 0.2 μm pore size membranes for final filtration of an oncolytic adenovirus. In general, the best performance was achieved using dual-layer sterilizing grade filters like the Fluorodyne EX EDF and the MiniSart Plus, which have a 0.45 μm layer directly on top of a 0.2 μm sterilizing grade filter. Similar results were obtained by Taylor et al. [115] during sterilizing grade filtration of a human cytomegalovirus live attenuated virus vaccine (LAV) candidate, with mean particle diameter from 100 - 400 nm. As shown in Figure 22, the filter capacity (defined as the volumetric throughput at a transmembrane pressure of 100 kPa) ranged from $<25 \text{ L/m}^2$ to more than 200 L/m^2 even though all of the membranes were rated as having 0.2 μm nominal pore size sufficient for complete retention of very small bacteria. Similar variability was seen in the LAV yield. In general, the multi-layer filters tended to out-perform single layer (asymmetric and homogeneous) filters, both in terms of capacity and LAV yield, although many of the multi-layer filters were unsuccessful in processing this LAV.

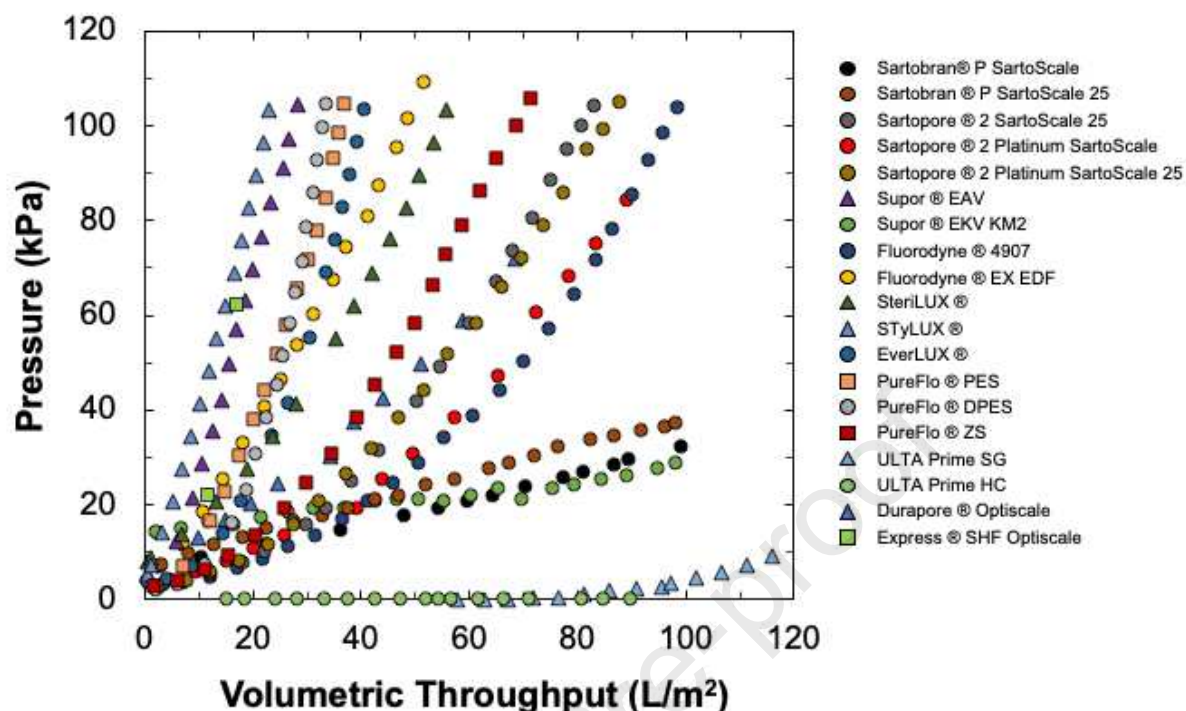


Figure 22: Transmembrane pressure profiles during filtration of an LAV product with approximately $3 \text{ to } 5 \times 10^{10}$ particles/mL through different sterilizing grade filters at a constant filtrate flux of $80 \mu\text{m/s}$ (300 LMH). Reproduced from [115].

Carvalho et al. [112] proposed a fully membrane-based process for purification of an influenza virus-like particle produce in insect cells. Initial clarification was performed by depth filtration through a double-layer cellulosic filter containing an appropriate filter aid followed by normal flow filtration through a $0.5 \mu\text{m} / 0.2 \mu\text{m}$ microfiltration membrane. Tangential flow filtration was then used to remove DNA and HCP, using first a 1000 kDa polyethersulfone membrane followed by a 300 kDa polyethersulfone membrane. The final product was then filtered through a $0.2 \mu\text{m}$ asymmetric polyethersulfone membrane. The overall virus recovery was 80% with approximately 30% reduction in process time and lower buffer consumption than a comparable chromatographic process, although the size-based membrane separation would be

ineffective at removing empty capsids. A similar membrane-based process was examined by Peixoto et al. [114] for the purification of an adenovirus vector for gene therapy, although in this case the process employed a Sartobind anion exchange membrane adsorber using a bind-and-elute operation with the elution performed using a simple salt gradient. The commercial viability of these processes is yet to be demonstrated.

4.4 Cell-based Therapies

Kymriah (developed by Novartis) was the first Chimeric Antigen Receptor (CAR) T-cell therapy approved by the FDA in August 2017 followed shortly thereafter by Yescarta (from Gilead) [117]. These therapeutics use the patient's own (autologous) cytotoxic T-lymphocytes, which are collected and genetically engineered to express an antigen-specific receptor that enables the cells to recognize and kill target tumor cells. The RNA sequence for the chimeric antigen receptor, which is a fusion protein containing the antigen binding domain of a monoclonal antibody and specific domains for T-cell signaling, is typically transferred to the T-cells using a lentivirus vector [118].

The production of a CAR-T therapeutic begins with collection of T-lymphocytes (white blood cells) from the patient's blood by selective centrifugation, often known as leukapheresis, with the T-cells collected in the fraction between the more dense red blood cells and the supernatant plasma. This is followed by T-cell activation, e.g., using antibody-coated magnetic beads (e.g., Dynabeads) or nanobeads, to facilitate the genetic modification [118]. The T-cells are then genetically-modified, expanded in an appropriate bioreactor, concentrated, and then formulated for cryopreservation (if needed) and delivery to the patient.

There is also growing interest in the use of human mesenchymal stem cells for the repair and / or regeneration of various tissues, e.g., cardiac tissue for treatment of heart disease and pancreatic tissue for the treatment of diabetes [119]. In addition, hematopoietic stems cells obtained from the bone marrow have been explored for the treatment of Crohn's disease [119].

There are a number of opportunities for using membrane systems in the production of cell-based therapeutics. The LOVO Cell Processing System from Fresenius Kabi was specifically designed for concentration and washing of harvested blood cells [120]. The LOVO is a pre-sterilized single-use microfiltration device employing a 4 μm pore size track-etch polycarbonate membrane with very uniform cylindrical pores that is capable of separating cells with different size. The membrane is wrapped around the inner cylinder of a spinning membrane filtration unit, in a configuration often referred to as a Taylor-Couette device based on the development of Taylor vortices in the narrow gap between the inner and outer cylinders (Figure 23). These Taylor vortices increase back mass transfer and help reduce fouling of the membrane. Schlinker [120] demonstrated that the LOVO was able to remove 99.8% of the platelets (about 2-3 μm in size) in the permeate with more than 90% recovery of CD34⁺ hematopoietic stem cells after a 2-step washing process. More traditional tangential flow filtration using 0.45 μm pore size hollow fiber polysulfone membranes has also been used for concentration of human mesenchymal stem cells [121].

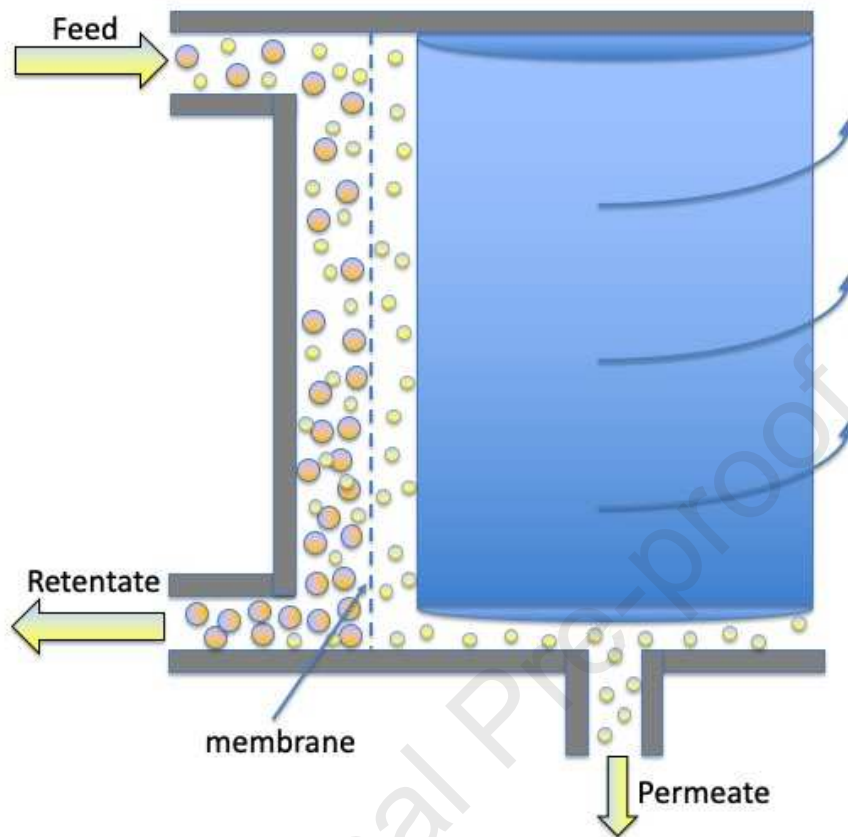


Figure 23: Schematic of the LOVO spinning membrane Cell Processing System. The smaller (yellow) cells represent platelets while the larger (orange) cells represent stem cells.

Normal flow filters with very large pore size (30-100 μm) have been successfully used for the recovery of human mesenchymal stem cells (hMSC) after detachment of the cells from the large microcarriers ($>125 \mu\text{m}$) used for adherent cell culture [121]. Nylon filters have been used to recover human adipose-derived stem cells from the harvested fat tissue based on the strong adsorption of these stem cells to the nylon membrane [122]. Cells could then be recovered by reverse flushing with fetal bovine serum. It was also possible to select for cells with high

ability to differentiate into bone cells using an affinity membrane in which the nylon was coated with a layer of recombinant vitronectin, a glycoprotein that promotes cell adhesion.

Membrane systems are also attractive for point of care removal of cryoprotective agents like glycerol and dimethyl sulfoxide (DMSO). TFF systems have been shown to provide very effective removal of glycerol from red blood cell suspensions [123], although the very small volumes needed for most cell-based therapies would be difficult to process using traditional TFF modules. Mfarrej et al. [124] demonstrated that the LOVO spinning membrane filter could effectively remove DMSO in the processing of CD34⁺ progenitor cells. Tostoes et al. [125] developed a novel membrane device that combines both tangential flow and normal flow filtration while exploiting cell settling to minimize membrane fouling (Figure 24). In this case, the 0.65 μm pore size PVDF membrane is kept largely free of cells, which are allowed to settle in the bottom collection chamber during the loading and washing stages. The cells can then be concentrated by pumping a small amount of air through the inlet to force fluid through the membrane. The system provided more than 85% removal of DMSO with high recovery of bone marrow-derived human mesenchymal stem cells.

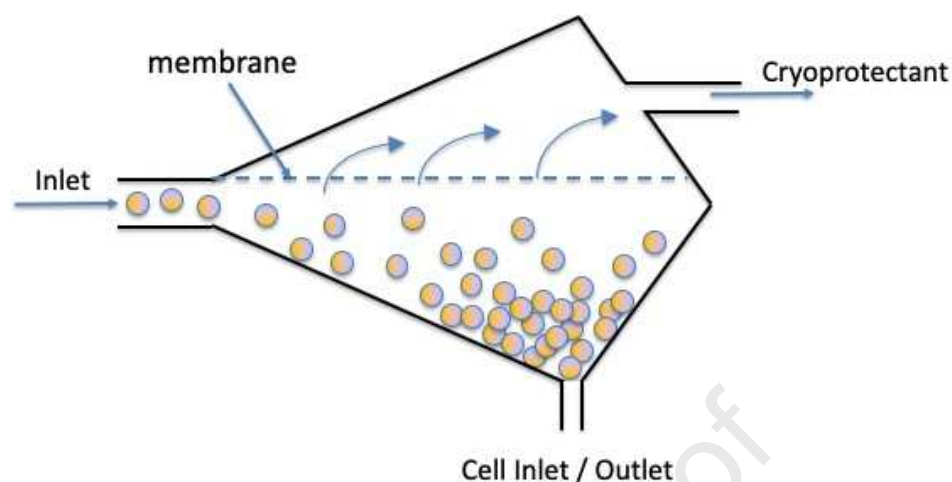


Figure 24: Schematic of the cell settling / filtration device for point-of-care washing of cell therapeutics.

It may also be possible to use membrane systems for cell fractionation based on differences in size and / or deformability. Fan et al. [126] developed a microfluidic device incorporating a 6-10 μm pore size polydimethylsiloxane (PDMS) membrane that was able to separate circulating tumor cells from blood, with the smaller red blood cells and lymphocytes washed through the membrane for removal in the permeate. Guzniczak et al. [127] used a dead-end filtration through 3 μm pore size track-etch membranes for the collection of enucleated (manufactured) red blood cells. The membrane device was used in combination with a spiral inertial separation, with the enucleated cells collected in the permeate since they are able to deform and pass through relatively small pores. Additional work will be required to demonstrate the viability of this type of deformability-based cell separation and its potential application in the purification / processing of novel cell therapies.

5. Conclusions

Recent advances in cell and molecular biology continue to spur the development of novel biotherapeutics for both the treatment and prevention of disease. This includes the use of gene therapy for the treatment of hereditary disorders, cell-based therapies for treatment of cancer and replacement / regeneration of tissues, vaccines for immunization against viral and bacterial infections, and a range of antibody-derived products (e.g., bi-specifics and antibody drug conjugates) for treatment of cancer and various immunological disorders. The successful commercialization of these novel biotherapeutics requires the development of effective biomanufacturing operations, and membrane technology is poised to play a critical role in this effort.

Much of the current work on downstream processing has focused on the application of existing membrane systems, often with small modifications, to address some of the key challenges associated with the production of these large / complex products. This includes identification of sterilizing grade filters that are suitable for processing viral vaccines, the development of membrane adsorbers with greater binding capacity, and the application of traditional ultrafiltration membranes for separation of plasmid DNA isoforms. In many cases, this also requires the development of improved process analytical technology (PAT) to facilitate scale-up and to ensure operation remains within the critical process parameters [128, 129].

However, there are numerous future opportunities that will likely require the development of novel membranes, modules, and / or processes that are able to meet the unique challenges of the bioprocessing industry. HPTFF could potentially become a viable technology for the commercial purification of several of these novel therapeutic modalities, providing separations that would be difficult to achieve using alternative technologies. There are also

exciting opportunities to develop hybrid membrane processes that exploit the benefits of membranes in combination with other separating agents like affinity precipitation. In addition, it may be possible to develop membranes that can achieve selective cell separations based on differences in cell size and deformability, membrane systems that can differentiate between filled and unfilled viral capsids, membrane modules that will enable continuous formulation of biotherapeutics, affinity membrane adsorbers with enhanced selectivity based on the presence of specific cell surface markers, and low-fouling membranes / modules for long-term operation of perfusion bioreactors. In many cases this would require substantive advances in membrane technology that are well beyond current capabilities. It will be exciting to see the membrane industry rise to meet these challenges in the years to come.

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Highlights

- Recent advances in bioprocessing have created new challenges for membranes
- Continuous bioprocessing requires new strategies for implementing membranes
- Single-use flexible manufacturing requires new membrane / module designs
- Purification of gene and cell-based therapies requires new separation technologies
- Future developments in bioprocessing will drive continued advances in membranes

Conflict of Interest

The author has no conflicts of interest to disclose.